



CCS370-UI AND UX DESIGN-LAB-MANUAL - III CSE & III AIDS

UI & UX design (Anna University)



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UNIT-1

FOUNDATIONS OF DESIGN

UNIT I FOUNDATIONS OF DESIGN

UI vs. UX Design - Core Stages of Design Thinking - Divergent and Convergent Thinking - Brainstorming and Game storming - Observational Empathy

UI Vs. UX DESIGN:**UX vs UI: what's the difference?**

“UX design” stands for “User Experience design.” UX teams study how users interact with a product or service to understand the users’ needs. These insights allow designers to create an intuitive, efficient product that people will actually want to use.

“UI” is short for “User Interface.” UI focuses on the screens, icons, buttons, visual elements and interfaces in a user experience.

What is UX design?

UX designers aim to create products with relevant, meaningful, usable and pleasurable experiences. They look at a product holistically and focus on meeting the users’ needs. UX designers use a process called “user-centered design.” The practice of UX design ensures that a product is easy to use and enjoyable. UX design practices could even make that experience pleasurable.

Interior designers and architects create physical spaces that are comfortable and easy to navigate; UX designers do the same for digital spaces. To make the best user experience, designers consider the context in which users will use the product (for instance, do they use it while on the move). They also design around the device the product appears on or what time of day the product gets the most usage.

Accessibility, information hierarchy and navigation flows are part of UX and require frequent collaboration with the UI designers to ensure the product works for the most users possible. The goal is to reduce “friction”—a term for unnecessary difficulty or stress when using a product.

What is UI design?

UI designers build interfaces for digital products or services centered on aesthetics and utility.

UI is more specialized than UX design because it exclusively focuses on the elements that users directly interact with, such as buttons and icons.

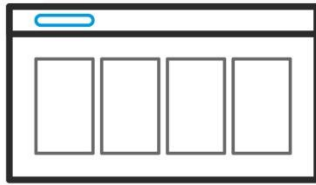
While UI elements tend to be visual, there are other types of UI, too—for example, voice- and gesture-based interfaces.

There’s often a misconception that UI design is virtually the same as graphic design.

While there are similarities and they apply some of the same skills, UI designers create interactive visual elements (buttons, screen animations, layouts), and graphic designers make static elements (logos, header images). A user experience contains several user interfaces that create a whole product.



Common Types of User Interface



Graphical User Interfaces (GUIs)



Voice-controlled Interfaces (VUIs)



Gesture-based Interfaces

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User interfaces are the most tangible aspects of a user's experience. If UI is right, and people can better navigate a product that is appealing and intuitive.

Examples of UX vs UI

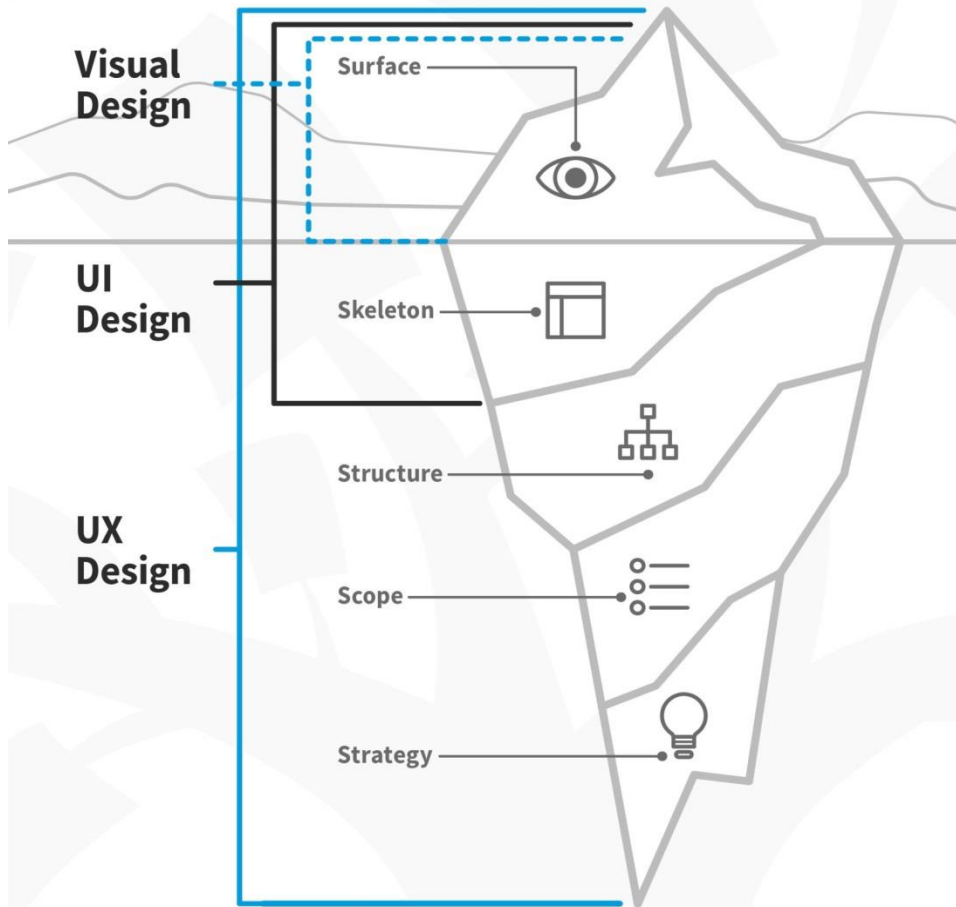
User experience is a user's overarching, all-embracing, holistic experience. It encompasses everything the user sees and feels, including the problem the product is trying to solve and where a product is used. User interface refers only to the screen, buttons or other things that make up the interface.

Example:

From the moment you step into a grocery store to the moment you walk out with your groceries, you're in the store's user experience. Everything you see is designed with purpose, from the aisle size to the organization of the shelves, even the temperature. All of this forms the user experience. If you've gone shopping at your usual store, but instead of paying via a cashier, you opt for self-checkout, the interface you use to ring up your items and pay is part of the user interface.

In summary, UX includes UI, and UI is part of a user experience.

User Interface Design



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Like an iceberg, visual design is only the surface of a user experience. Underneath, there is much more, including a skeleton, structure, scope and strategy. These layers influence each other, so are by no means

independent. Decisions taken on one plane can affect other layers. New issues or opportunities could arise, which might impact the experience. For example, if the team encounters technical challenges or budgetary constraints during development, they might have to revisit some design decisions.

If you have the store's app on your phone or visit its website, that is part of the user experience. To fully enjoy that experience, the user interacts with the interface.

UX design decides what features are going to be in the app.

Will it have a loyalty program where you gain points for each purchase?

Will the user be able to track the points on the app?

Perhaps the app allows you to shop online or see the current specials on offer—all of that is part of the user experience.

However, the typeface, layout, icons, spacing and visual style are all decided by user interface design.

UX is more abstract. It incorporates a user's environment, mood and context.

UI, on the other hand, is more tangible—you directly interact with the UI.

Differences Between UX And UI Design:

The table below shows the most significant differences between UX and UI design:

UX Design	UI Design
Focuses on the holistic experience of the user	Focuses on the specific visual touch points of the user
Centers on strategy, structure and interaction design	Centers on surface-level aspects of design, including visuals
Involves studying the user's journey and designing information architecture	Involves designing the tangible elements of the experience such as visual style, e.g., color palettes, typography and layout
Outputs include personas, user journey maps, wireframes, prototypes	Outputs include mockups, high-fidelity layouts, animations and imagery
The goal of UX design is to identify and solve user problems.	The goal of UI is to create attractive, interactive and intuitive interfaces.

Generally, UX designers handle the initial steps in the product development process, followed by UI. The UX designer maps out the skeleton of a user journey. Then the UI designer creates the visual and interactive elements.

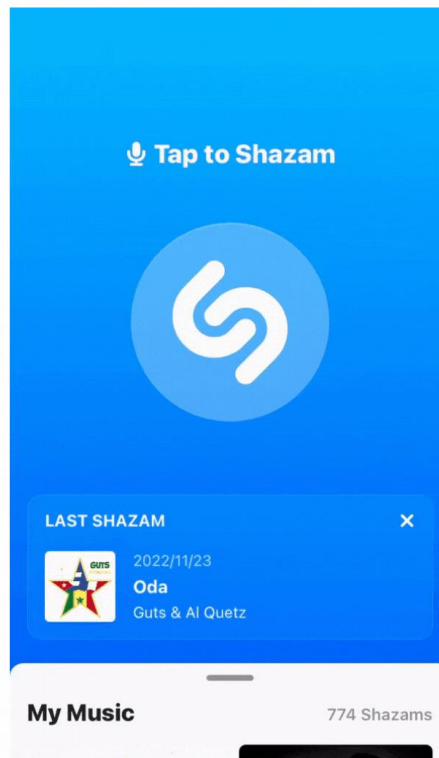
UX and UI Design: How Do They Work Together?

UX and UI overlap but are distinct disciplines. UX is the umbrella, and UI falls under it. Both are needed for a product to succeed—if you've got a website with an attractive interface but it's hard to use and navigate (a bad user experience), the website won't succeed. An old-fashioned, unappealing website that's straightforward and easy to use is not as much of an issue, but it can still affect a user's enjoyment of the product. They might prefer a competitor's product, one that's just as usable but much more aesthetically pleasing.

To illustrate how UX and UI work together, let's look at Shazam. Shazam identifies music playing around you. This app solves a real-life problem: how often have you heard a great song on the radio, at a shop or bar

but couldn't identify it? This user experience solves this specific problem. Now you can not only identify the song, but you can also store that song for future reference. Shazam connects to music streaming apps like Apple Music or Spotify and creates a playlist of all the tracks you have 'Shazamed.' The UI is simple and clean because that's what the user requires.

Imagine you've just turned on the radio, and the song that's playing is trailing off, coming to an end—you have just a few seconds to get your phone out. The app caters to this exact issue, as the interface is pared back, with just one large button on the screen. That button is animated to signal to the user to press it, with a line of text saying, "Tap to Shazam."



The animation, the icon and the text are all part of UI design and add to the positive user experience. It demonstrates how UX and UI can and should work together.

UX vs UI Design: Which Career Path Should be Taken?

UX and UI are intertwined. That's why so many job ads are for a UX/UI designer. If you're doing UX design, it would be helpful to have some UI skills, especially for projects with low time and resources. In an ideal world, however, the jobs would be separate, and the UX designer and UI designer would collaborate.

Having UX and UI design skills is beneficial, but you don't have to master both. Decide which path you would like to take and focus on building your skills and experience in that area. You'll likely pick up something about the other as they're interconnected. UX and UI design are in high demand and are generally well-paid, so this is the opportune moment to investigate a career in either of these disciplines.

Let's look at what you would be responsible for in each role to put yourself in the best position to decide which path you'd like to take. See our responsibility comparison below.



Responsibility Comparison

UI Designer:

- Collaborating with product management and engineering to define and implement innovative solutions for the product direction, visuals and experience
- Executing all visual design stages from concept to final hand-off to engineering
- Conceptualizing original ideas that bring simplicity and user friendliness to complex design roadblocks

UI/UX Designer:

- Gathering and evaluating user requirements, in collaboration with product managers and engineers
- Illustrating design ideas using storyboards, process flows and sitemaps
- Designing graphic user interface elements, like menus, tabs and widgets

UX Designer:

- Conducting user research and testing
- Developing wireframes and task flows based on user needs
- Collaborating with Designers and Developers to create intuitive, user-friendly software

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The soft skills you need for both roles are virtually the same and would be beneficial in both jobs. The hard skills are more distinct for each role. This list is not exhaustive but highlights some of the key soft and hard skills for UX and UI design.

UX Design

Soft skills	Hard skills
Empathy	Prototyping
Communication	User research techniques
Strategy	Analysis
Problem-solving	Wireframing
Collaboration	Information architecture

UI Design

Soft skills	Hard skills
Empathy	Prototyping
Communication	Animation
Creativity	Color theory
Adaptability	Typography
Collaboration	Design patterns

Part of what makes UX and UI design so appealing is there are so many transferable skills. Some of them are:

- Graphic design
- Project management
- Research
- Marketing
- Customer service

What Does a UX Designer Actually Do?

UX design is varied and multi-disciplinary. Conduct user research through interviews, usability tests and card sorting (among other methods) to discover user behavior, needs and pain points (the users' problems). To develop products, you'll conduct competitor analyses and craft product strategies. Develop content, wireframes and prototypes.

What Does a UI Designer Actually Do?

UI involves in design's visual and creative aspects. In terms of aesthetics, Develop a visual design, including graphics and typography. UI focus on the interactivity of a project. UI ensure layouts work well, look good on various devices, and create animations and UI prototypes. UI designer will work closely with a developer so that your designs come to life exactly as you envisioned them.

CORE STAGES OF DESIGN THINKING

What is design thinking?

Definition:

Design thinking is a non-linear, iterative process that teams use to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test.

Design Thinking



Empathize



Define



Ideate



Prototype



Test

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Design thinking is a methodology which provides a solution-based approach to solving problems. It's extremely useful when used to tackle complex problems that are ill-defined or unknown—because it serves to understand the human needs involved, reframe the problem in human-centric ways, create numerous ideas in brainstorming sessions and adopt a hands-on approach to prototyping and testing.

Why is design thinking so important?

*In user experience (UX) design, it's crucial to develop and refine skills to understand and address rapid changes in users' environments and behaviors

*Design teams use design thinking to tackle ill-defined/unknown problems (aka wicked problems) because they can reframe these in human-centric ways and focus on what's most important for users.

*Of all design processes, design thinking is almost certainly the best for "thinking outside the box". With it, teams can do better UX research, prototyping and usability testing to uncover new ways to meet users' needs.

*Design thinking's value as a world-improving, driving force in business (global heavyweights such as Google, Apple and Airbnb have wielded it to notable effect) matches its status as a popular subject at leading international universities.

*With design thinking, teams have the freedom to generate ground-breaking solutions.

*Using it, your team can get behind hard-to-access insights and apply a collection of hands-on methods to help find innovative answers.

In essence, design thinking:

Revolves around a deep interest to understand the people for whom we design products and services.

Helps us observe and develop empathy with the target users.

Enhances our ability to question: in design thinking you question the problem, the assumptions and the implications.

Proves extremely useful when you tackle problems that are ill-defined or unknown.

Involves ongoing experimentation through sketches, prototypes, testing and trials of new concepts and ideas.

Design Thinking Makes You Think Outside the Box

Design thinking can help people do out-of-the-box or outside-the-box thinking. People who use this methodology:

Attempt to develop new ways of thinking—ways that do not abide by the dominant or more common problem-solving methods.

Have the intention to improve products, services and processes. They seek to analyze and understand how users interact with products to investigate the conditions in which they operate.

Ask significant questions and challenge assumptions. One element of outside-the-box / out-of-the-box thinking is to falsify previous assumptions—i.e., make it possible to prove whether they're valid or not.

As you can see, design thinking offers us a means to think outside the box and also dig that bit deeper into problem-solving. It helps us carry out the right kind of research, create prototypes and test our products and

services to uncover new ways to meet our users' needs.

Design Thinking is for Everybody

Design thinking is not only for designers but also for creative employees, freelancers and leaders who seek to infuse it into every level of an organization. This widespread adoption of design thinking will drive the creation of alternative products and services for both business and society.

The process is firmly based on how you can generate a holistic and empathic understanding of the problems people face. Design thinking involves ambiguous, and inherently subjective, concepts such as emotions, needs, motivations and drivers of behavior.

In a solely scientific approach (for example, analyzing data), people are reduced to representative numbers, devoid of emotions. Design thinking, on the other hand, considers both quantitative as well as qualitative dimensions to gain a more complete understanding of user needs. For example, you might observe people performing a task such as shopping for groceries, and you might talk to a few shoppers who feel frustrated with the checkout process at the store (qualitative data). You can also ask them how many times a week they go shopping or feel a certain way at the checkout counter (quantitative data). You can then combine these data points to paint a holistic picture of user pain points, needs and problems.

Design Thinking Has a Scientific Side

Design thinking is both an art and a science. It combines investigations into ambiguous elements of the problem with rational and analytical research—the scientific side in other words. This magical concoction reveals previously unknown parameters and helps to uncover alternative strategies which lead to truly innovative solutions.

The scientific activities analyze how users interact with products, and investigate the conditions in which they operate. They include tasks which:

Research users' needs.

Pool experience from previous projects.

Consider present and future conditions specific to the product.

Test the parameters of the problem.

Test the practical application of alternative problem solutions.

Once you arrive at a number of potential solutions, the selection process is then underpinned by rationality. As a designer, you are encouraged to analyze and falsify these solutions to arrive at the best available option for each problem or obstacle identified during phases of the design process.

With this in mind, it may be more correct to say design thinking is not about thinking outside the box, but on its edge, its corner, its flap, and under its bar code—as Clint Runge put it.

Resetting Our Mental Boxes and Developing a Fresh Mindset

Thinking outside of the box can provide an innovative solution to a sticky problem. However, thinking outside of the box can be a real challenge as we naturally develop patterns of thinking that are modeled on the repetitive activities and commonly accessed knowledge we surround ourselves with.

The Design Thinking Process

Design thinking is essentially a process which moves from problem to solution via some clear intermediate points. The classic approach, as proposed by Herbert A Simon, is offered here:

Definition – where the problem is defined as best as possible prior to solving it

Research – where the designers examine as much data as they feel necessary to be able to fully contribute to the problem solving process

Ideation – where the designer commences creating possible solutions without examining their practicality until a large number of solutions has been proposed. Once this is done, impractical solutions are eliminated or played with until they become practical.

Prototyping – where the best ideas are simulated in some means so that their value can be explored with users

Choosing – where the best idea is selected from the multiple prototypes

Implementing – where that idea is built and delivered as a product

Testing – where the product is tested with the user in order to ensure that it solves the original problem in an effective manner

There are many other design thinking processes outlined in literature – most of which are a truncated version of the above process combining or skipping stages.

The Principles of Design Thinking

Human – all design is of a social nature

Ambiguity – design thinking preserves and embraces ambiguity

Re-design – all design processes are in fact re-design of existing processes

Tangibility – the design process to make something tangible will facilitate communication of that design

Core stages of design thinking:

We focus on the five-stage design thinking model proposed by the Hasso Plattner Institute of Design at Stanford (the d.school) because they are world-renowned for the way they teach and apply design thinking.

The five stages of design thinking, according to the d.school, are:

Empathize: research your users' needs.

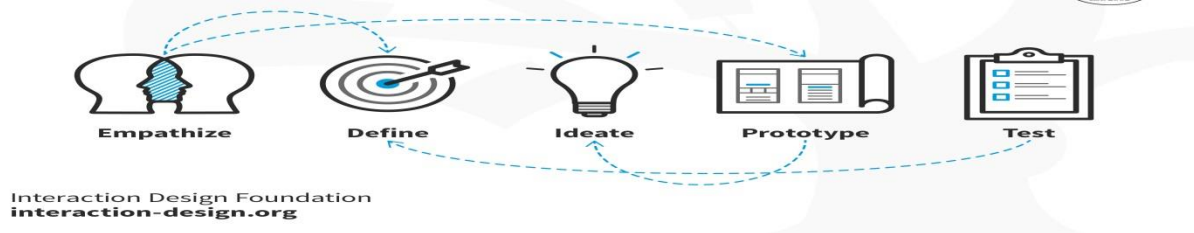
Define: state your users' needs and problems.

Ideate: challenge assumptions and create ideas.

Prototype: start to create solutions.

Test: try your solutions out.

Design Thinking: A 5-Stage Process



Design thinking is an iterative and non-linear process that contains five phases: 1. Empathize, 2. Define, 3. Ideate, 4. Prototype and 5. Test.

Stage 1: Empathize—Research Your Users' Needs

Empathize



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Empathize: the first phase of design thinking, where you gain real insight into users and their needs.

The first stage of the design thinking process focuses on user-centric research. It is needed to gain an empathic understanding of the problem you are trying to solve. Consult experts to find out more about the area of concern and conduct observations to engage and empathize with your users. You may also want to immerse yourself in your users' physical environment to gain a deeper, personal understanding of the issues involved—as well as their experiences and motivations. Empathy is crucial to problem solving and a human-centered design process as it allows design thinkers to set aside their own assumptions about the world and gain real insight into users and their needs.

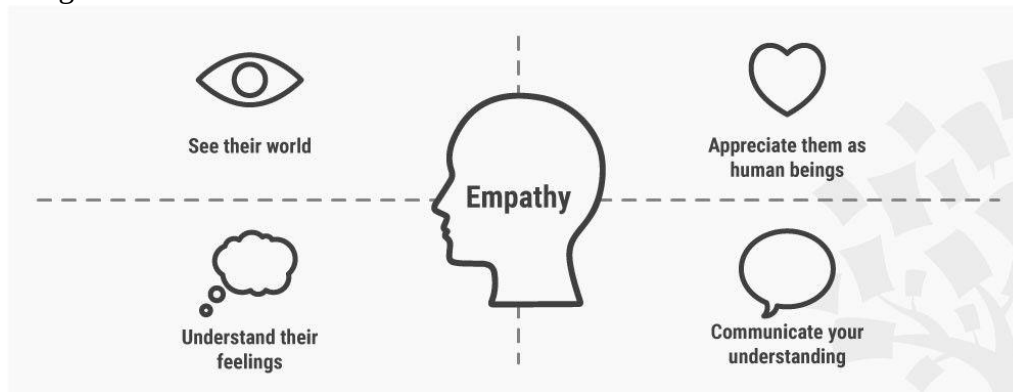
Depending on time constraints, you will gather a substantial amount of information to use during the next stage. The main aim of the Empathize stage is to develop the best possible understanding of your users, their needs and the problems that underlie the development of the product or service you want to create.

Empathize with Real People – Leave Your Assumptions Outside

Empathize is design thinking's first stage for a reason. It's the first step on the road to thoughtfully designed products that prove the designers built with a compassionate eye for their users. Empathy is a naturally occurring characteristic which people have in varying degrees. However, they can improve their ability to empathize as a soft skill. Anyone in a design team will have preconceived ideas about the many situations people find themselves in as users. It's unavoidable – you can't unlearn your life experience. Therefore, you should always adopt a beginner's mindset to be able to view and analyze situations with users objectively.

To empathize is to research. So, you should constantly remind yourself to question everything you observe instead of judging. You should also listen to others open-mindedly rather than focus on points that confirm your biases. Because our biases will naturally creep into how we view the world and the situations we

consider, as designers—or design thinkers—we must catch and overcome these before they distort our research. You must become fully objective before you can start to see through your users’ eyes and interpret their viewpoints optimally. They are the experts. You must understand the users’ dimensions of use (e.g., tasks) and their feelings (e.g., motivations) before you can work towards delighting them through your design.



How to Empathize to Get the Right Insights

You have a range of options, including:

Observing real users. Ask these questions to shift from concrete observations to abstract motivations:

“What?” – You detail your observations.

“How?” – You analyze how users do things (e.g., with difficulty).

“Why?” – You make educated guesses about the users’ emotions and motivations.

Conducting photo- and/or video-based studies in users’ natural environments or sessions with the design team or consultants – You record these users while they try to solve an issue you propose to resolve with your design.

Personal photo/video journals – You ask users to record their own experiences with approaching a problem. These may capture their pain points more accurately.

Interviewing users – Your team uses brainstorming to first find the right questions to ask in a generally structured and natural flow. Then, you can directly ask users for their insights in an intimate setting where they can respond earnestly to open-ended questions.

Engaging with extreme users – You find the extreme cases within your userbase to determine the greatest degrees of users’ needs, problems and problem-solving methods. You can then see the full scope of problems which typical, non-extreme users might run into. If you can satisfy an extreme user, you can satisfy any user.

Analogous empathy – Your team finds effective analogies to draw parallels between users’ problems and problems in other fields. This way, you can get insights you’d otherwise overlook.

Sharing inspiring stories – Your team shares stories about what they have observed so you draw meaning from these and note fascinating details.

Bodystorming – You wear equipment (e.g., goggles, gloves, torso attachments) to gain first-hand experience of your users in their environment.

Empathy maps and customer journey maps – Your team should have at least one of these as a reference point to appreciate the users’ perspectives.

Personas to establish accurate portraits/profiles of users who’ll interact with your product.

Whichever approaches you take, beware of formulating solutions at this stage. Aim to realistically envision possible scenarios where users experience problems. Empathize is not just a key part of design thinking. It’s

also pivotal to user-centered design and user experience (UX) design. When your design team remains aware of your users' realities and passionate about helping real people solve real problems, you'll reap precious insights which you can ultimately translate into products your users will love.

Stage 2: Define—State Your Users' Needs and Problems

Define

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Define: the second phase of design thinking, where you define the problem statement in a human-centered manner.

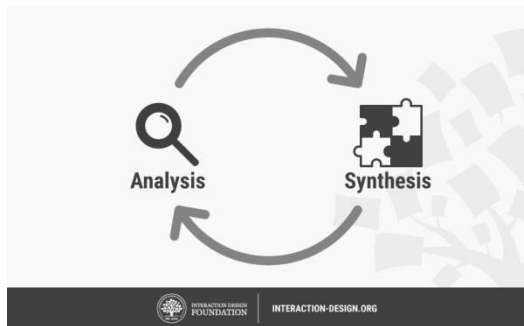
In the Define stage, you will organize the information you have gathered during the Empathize stage. You'll analyze your observations to define the core problems you and your team have identified up to this point. Defining the problem and problem statement must be done in a human-centered manner.

For example, you should not define the problem as your own wish or need of the company: "We need to increase our food-product market share among young teenage girls by 5%."

You should pitch the problem statement from your perception of the users' needs: "Teenage girls need to eat nutritious food in order to thrive, be healthy and grow."

The Define stage will help the design team collect great ideas to establish features, functions and other elements to solve the problem at hand—or, at the very least, allow real users to resolve issues themselves with minimal difficulty. In this stage, you will start to progress to the third stage, the ideation phase, where you ask questions to help you look for solutions: "How might we encourage teenage girls to perform an action that benefits them and also involves your company's food-related product or service?" for instance.

In the Define stage you synthesise your observations about your users from the first stage, the Empathise stage. A great definition of your problem statement will guide you and your team's work and kick start the ideation process (third stage) in the right direction. The five stages are not always sequential — they do not have to follow any specific order and they can often occur in parallel and be repeated iteratively. As such, the stages should be understood as different modes that contribute to a project, rather than sequential steps.



Before we go into what makes a great problem statement, it's useful to first gain an understanding of the relationship between analysis and synthesis that many design thinkers will go through in their projects.

Analysis is about breaking down complex concepts and problems into smaller, easier-to-understand constituents. We do that, for instance, during the first stage of the Design Thinking process, the Empathise stage, when we observe and document details that relate to our users. Synthesis, on the other hand, involves creatively piecing the puzzle together to form whole ideas. This happens during the Define stage when we organise, interpret, and make sense of the data we have gathered to create a problem statement.

What are Problem Statements?

Problem statements are concise descriptions of design problems. Design teams use them to define the current and ideal states, and to freely find user-centered solutions. Then, they use these statements—also called points of view (POVs)—as reference points throughout a project to measure the relevance of ideas they produce.

How to Write a Problem Statement?

Well-constructed, valid and effective problem statements are vital for your design team to navigate the entire design process. Essential to design thinking, problem statements are what teams produce in the Define stage. To find the best solutions, your team must know what the exact problems are—i.e., you first need to define a problem statement.

The goal is to articulate the problem so everyone can see its dimensions and feel inspired to systematically hunt for suitable solutions. When you unite around a problem statement, your team will have a common view of how users see what they must tackle. From there, all your team will know exactly what to look for and what to avoid. Therefore, you should make your problem statements:

Human-centered: Frame problem statements from insights about users and their needs.

Have the right scope:

Broad enough to permit creative freedom, so you don't concentrate too narrowly on specific methods for implementing solutions or describing technical needs; but

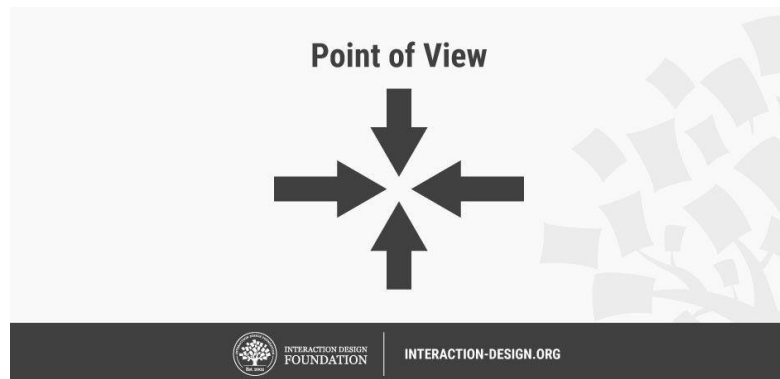
Narrow enough to be practicable, so you can eventually find specific solutions.

Based on an action-oriented verb (e.g., "create" or "adapt").

Fully developed and assumption-free.

Design teams sometimes refer to a problem statement as a “point of view” (POV) because they should word problem statements from the users’ perspective and not let bias influence them. Your team will have a POV when it comes up with a narrowly focused definition of the right challenge to pursue in the next stage of the design process.

With an effective POV, your team can approach the right problem in the right way. Therefore, you’ll be able to seek the solutions your users want.

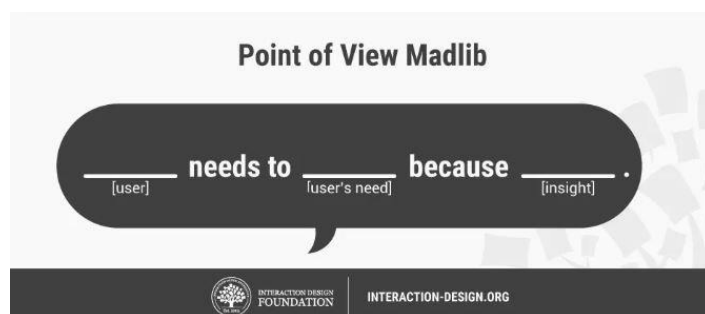


How to Define Problem Statements through a Point of View Madlib

To define a problem statement, your team must first examine recorded observations about users. You must capture your users’ exact profile in the problem statement or POV. So, you need to synthesize research results and produce insights that form solid foundations. From these, you can discover what those specific users really require and desire—and therefore ideate effectively.

Teams typically use a POV Madlib to reframe the challenge meaningfully into an actionable problem statement. The POV madlib is a framework you use to place the user, need and insight in the best way. This is the format to follow:

[User... (descriptive)] needs [need ... (verb)] because [insight... (compelling).]



You articulate a POV by combining these three elements—user, need, and insight—as an actionable problem statement that will drive the rest of your design work. Find an example below.

Point of View Template – Example

User	Need	Insight
An adult person who lives in a city	To use a car for 10-60 minute trips 1-4 times per week	The user would not want to own his own car as it would be too expensive compared to his needs. He would like to share a car with others who have similar needs, however, there are no easy and affordable solutions for him. It's important for the user to think and live green and to not own more than he truly needs.



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With a valid problem statement, your team can explore the framed “why” questions with “how”-oriented ones. That’s how you proceed to find potential solutions. You’ll know you have a good problem statement if team members:

Feel inspired.

Have the criteria to evaluate ideas.

Can use it to guide innovation efforts.

Can’t find a cause or a proposed solution in it (which would otherwise get in the way of proper ideation).

When your team has a good problem statement, everyone can compare ideas, which is vital in brainstorming and other ideation sessions. It also means everyone can keep on the right track. Problem statements are powerful aids because they encourage well-channeled divergent thinking.

Rather than rush toward solutions that look impressive but aren’t effective, your team can work imaginatively to find the right ones. Once you’ve discovered what’s causing problems, you can give users the best solutions in designs they like using.

What Makes a Good Problem Statement?

A problem statement is important to a Design Thinking project, because it will guide you and your team and provides a focus on the specific needs that you have uncovered. It also creates a sense of possibility and optimism that allows team members to spark off ideas in the Ideation stage, which is the third and following stage in the Design Thinking process. A good problem statement should thus have the following traits. It should be:

Human-centered. This requires you to frame your problem statement according to specific users, their needs

and the insights that your team has gained in the Empathise phase. The problem statement should be about the people the team is trying to help, rather than focusing on technology, monetary returns or product specifications.

Broad enough for creative freedom. This means that the problem statement should not focus too narrowly on a specific method regarding the implementation of the solution. The problem statement should also not list technical requirements, as this would unnecessarily restrict the team and prevent them from exploring areas that might bring unexpected value and insight to the project.

Narrow enough to make it manageable. On the other hand, a problem statement such as , “Improve the human condition,” is too broad and will likely cause team members to easily feel daunted. Problem statements should have sufficient constraints to make the project manageable.

As well as the three traits mentioned above, it also helps to begin the problem statement with a verb, such as “Create”, “Define”, and “Adapt”, to make the problem become more action-oriented.

How to Define a Problem Statement

Methods of interpreting results and findings from the observation oriented Empathise phase include:

Space Saturate and Group and Affinity Diagrams – Clustering and Bundling Ideas and Facts

In space saturate and group, designers collate their observations and findings into one place, to create a collage of experiences, thoughts, insights, and stories. The term 'saturate' describes the way in which the entire team covers or saturates the display with their collective images, notes, observations, data, experiences, interviews, thoughts, insights, and stories in order to create a wall of information to inform the problem-defining process. It will then be possible to draw connections between these individual elements, or nodes, to connect the dots, and to develop new and deeper insights, which help define the problem(s) and develop potential solutions. In other words: go from analysis to synthesis.

When you’ve defined your design challenge in a POV, you can start to generate ideas to solve your design challenge. You can start using your POV by asking a specific question starting with: “How Might We” or “in what ways might we”. How Might We (HMW) questions are questions that have the potential to spark ideation sessions such as brainstorm. They should be broad enough for a wide range of solutions, but narrow enough that specific solutions can be created for them. “How Might We” questions should be based on the observations you’ve gathered in the Empathise stage of the Design Thinking process.

For example, you have observed that youths tend not to watch TV programs on the TV at home, some questions which can guide and spark your ideation session could be:

How might we make TV more social, so youths feel more engaged?

How might we enable TV programs to be watched anywhere, at anytime?

How might we make watching TV at home more exciting?

The HMW questions open up to Ideation sessions where you explore ideas, which can help you solve your design challenge in an innovative way.

Why-How Laddering

For this reason, during the Define stage designers seek to define the problem, and will generally ask why.

Designers will use why to progress to the top of the so-called Why-How Ladder where the ultimate aim is to find out how you can solve one or more problems. Your How Might We questions will help you move from the Define stage and into the next stage in Design Thinking, the Ideation stage, where you start looking for specific innovative solutions. In other words you could say that the Why-How Laddering starts with asking Why to work out How they can solve the specific problem or design challenge.

Stage 3: Ideate—Challenge Assumptions and Create Ideas

Ideate



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Ideate: the third phase of design thinking, where you identify innovative solutions to the problem statement you've created.

During the third stage of the design thinking process, designers are ready to generate ideas. You've grown to understand your users and their needs in the Empathize stage, and you've analyzed your observations in the Define stage to create a user centric problem statement. With this solid background, you and your team members can start to look at the problem from different perspectives and ideate innovative solutions to your problem statement.

There are hundreds of ideation techniques you can use—such as Brainstorm, Brainwrite, Worst Possible Idea and SCAMPER. Brainstorm and Worst Possible Idea techniques are typically used at the start of the ideation stage to stimulate free thinking and expand the problem space. This allows you to generate as many ideas as possible at the start of ideation. You should pick other ideation techniques towards the end of this stage to help you investigate and test your ideas, and choose the best ones to move forward with—either because they seem to solve the problem or provide the elements required to circumvent it.

Ideation Will Help You:

Ask the right questions and innovate.

Step beyond the obvious solutions and therefore increase the innovation potential of your solution.

Bring together perspectives and strengths of team members.

Uncover unexpected areas of innovation.

Create volume and variety in your innovation options.

Get obvious solutions out of your heads, and drive your team beyond them.

Ideation Methods to Spark Innovative Ideas

There are hundreds of ideation methods. Some methods are merely renamed or slightly adapted versions of more foundational techniques. Here you'll get brief overview of some of the best methods:

Brainstorm

Braindump

Brainwrite

Brainwalk

Challenge Assumptions

SCAMPER

Mindmap

Sketch or Sketchstorm

Storyboard

Analogies

Provocation

Movement

Bodystorm

Gamestorming

Cheatstorm

Crowdstorm

Co-Creation Workshops

Prototype

Creative Pause

Active Facilitation

Although many of us may have previously participated in a Brainstorm session, it is not always easy to facilitate a truly fruitful ideation session, which may be the reason why many of us have had negative experiences in the past. However, Ideation sessions can indeed be fun and exciting, but they demand a lot of preparation and team member concentration in order to be fruitful. To sit the team down with a blank piece

of paper and ask them to come up with ideas will likely result in failure. Likewise, to have everyone shout out their own ideas is likely to result in failure.

People need guidance, inspiration and activities, in a physical and cognitive manner, in order to get the process started. Ideation is a creative and concentrated process; those involved should be provided with an environment that facilitates free, open, and the non-judgemental sharing of ideas.

In Ideation sessions, it's important to create the right type of environment to help create a creative work culture with a curious, courageous, and concentrated atmosphere. Instead of using a boardroom with the CEO sitting at the head of the table, Design Thinking and Ideation sessions require a space in which everyone is equal. The Ideation room must have sufficient space for people to feel comfortable, but the atmosphere shouldn't be sterile, and team members shouldn't have to shout in order to be heard. You should also designate someone to take down contributors' ideas and draw/write them on the whiteboard/wall/poster. If the process begins to slow down and people seem to be running into a dead-end, the facilitator should impose constraints, such as: "what if there was no top-level navigation bar?" or "How-might-we go about the task if we were 8 years old?" Alternatively, you might want to set targets, such as filling a brainstorming sheet within ten minutes. To start understanding what it takes to facilitate a successful Ideation session, we'll take a closer look at the best Brainstorming rules.

Ideation Methods to Select Ideas

Once the Ideation session is complete, the ideas must be collected, categorized, refined, and narrowed down, so the team is able to select the best solutions, ideas, and strategies from a shortlist. These methods can help you select the best idea at the end of an Ideation session:

Post-it Voting or Dot Voting.

Four Categories Method

Bingo Selection

Idea Affinity Maps

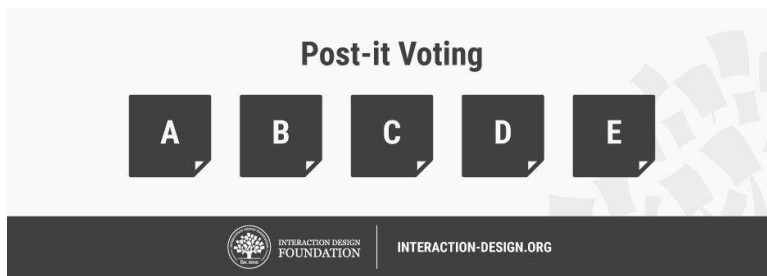
Now Wow How Matrix

Six Thinking Hats

Lean Startup Machine Idea Validation Board

Idea Selection Criteria

In the following section, we'll provide you with a brief introduction to some of the best methods.



Stage 4: Prototype—Start to Create Solutions

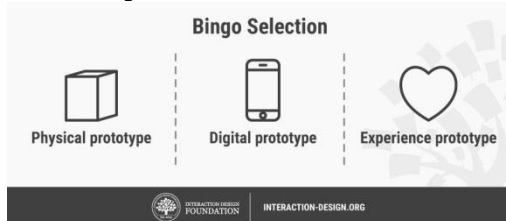
Prototype



In post-it voting, all members are given a number of votes (three to four should do) in order to choose their favorite ideas. Ideas that are generated in the Ideation sessions are written down on individual post-its, and members can vote by using stickers or a marker to make a dot on the post-it note corresponding to the ideas they like. This process allows every member to have an equal say in choosing from the shortlisted ideas.



The four categories method involves dividing ideas according to their relative abstractness, ranging from the most rational choice to the 'long shot' choice. The four categories are the rational choice, the most likely to delight, the darling, and the long shot. Members then decide upon one or two ideas for each of these categories. This method ensures that the team covers all grounds, from the most practical to those ideas with the most potential to deliver innovative solutions.



Prototype: the fourth phase of design thinking, where you identify the best possible solution.

The design team will now produce a number of inexpensive, scaled down versions of the product (or specific features found within the product) to investigate the key solutions generated in the ideation phase. These prototypes can be shared and tested within the team itself, in other departments or on a small group of people outside the design team.

This is an experimental phase, and the aim is to identify the best possible solution for each of the problems identified during the first three stages. The solutions are implemented within the prototypes and, one by one, they are investigated and then accepted, improved or rejected based on the users' experiences.

By the end of the Prototype stage, the design team will have a better idea of the product's limitations and the problems it faces. They'll also have a clearer view of how real users would behave, think and feel when they interact with the end product.

Types of Prototyping

Prototyping methods are generally divided into two separate categories: low- and high-fidelity prototyping.

Low-Fidelity Prototyping



Low-fidelity prototyping involves the use of basic models or examples of the product being tested. For example, the model might be incomplete and use just a few of the features that will be available in the final design, or it might be constructed using materials not intended for the finished article, such as wood, paper, or metal for a plastic product. Low-fidelity prototypes can either be models that are cheaply and easily made, or simply recounts or visualizations of them.

Examples of low-fidelity prototypes:

Storyboarding.

Sketching (although Bill Buxton, a pioneer of human-computer interaction, argues sketching is not an example of prototyping).

Card sorting.

'Wizard of Oz'.

Pros of Low-Fidelity Prototyping

Quick and inexpensive.

Possible to make instant changes and test new iterations.

Disposable/throw-away.

Enables the designer to gain an overall view of the product using minimal time and effort, as opposed to focusing on the finer details over the course of slow, incremental changes.

Available to all; regardless of ability and experience, we are able to produce rudimentary versions of products in order to test users or canvas the opinions of stakeholders.

Encourages and fosters design thinking.

Cons of Low-Fidelity Prototyping

An inherent lack of realism. Due to the basic and sometimes sketchy nature of low-fi prototypes, the applicability of results generated by tests involving simple early versions of a product may lack validity.

Depending on your product, the production of low-fi prototypes may not be appropriate for your intended users. For instance, if you are developing a product bound by a number of contextual and/or dispositional constraints (i.e., physical characteristics of your user base, such as users with disabilities), then basic versions that do not reflect the nature, appearance, or feel of the finished product may be of scant use; revealing very little of the eventual user experience.

Such prototypes often remove control from the user, as they generally have to interact in basic ways or simply inform an evaluator, demonstrate, or write a blow-by-blow account of how they would use the finished product.

High-Fidelity Prototyping



High-fidelity prototypes are prototypes that look and operate closer to the finished product. For example, a 3D plastic model with movable parts (allowing users to manipulate and interact with a device in the same manner as the final design) is high-fi in comparison to, say, a wooden block. Likewise, an early version of a software system developed using a design program such as Sketch or Adobe Illustrator is high-fi in comparison to a paper prototype.

Pros of High-Fidelity Prototyping

Engaging: the stakeholders can instantly see their vision realized and will be able to judge how well it meets their expectations, wants, and needs.

User testing involving high-fi prototypes will allow the evaluators to gather information with a high level of validity and applicability. The closer the prototype is to the finished product, the more confidence the design team will have in how people will respond to, interact with, and perceive the design.

Cons of High-Fidelity Prototyping

They generally take much longer to produce than low-fi prototypes.

When testing prototypes, test users are more inclined to focus and comment on superficial characteristics, as opposed to the content (Rogers, Preece, and Sharp, 2011).

After devoting hours and hours of time producing an accurate model of how a product will appear and behave, designers are often loathed to make changes.

Software prototypes may give test users a false impression of how good the finished article may be.

Making changes to prototypes can take a long time, thus delaying the entire project in the process. However, low-fi prototypes can usually be changed within hours, if not minutes, for example, sketching or paper prototyping methods are utilized.

Due to the pros and cons of low-fi and high-fi prototyping, it should be no surprise that low-fi prototyping is the usual option during the early stages of a Design Thinking project, while high-fi prototyping is used during the later stages, when the test questions are more refined.

Guidelines for Prototyping

It is important to remember that prototypes are supposed to be quick and easy tests of design solutions. Here are a few guidelines that will help you in the Prototyping stage:

Just start building

Design Thinking has a bias towards action: which means if you have any uncertainties about what you are trying to achieve, your best bet is to just make something. Creating a prototype will help you think about your idea in a concrete manner, and potentially allow you to gain insights into ways you can improve it.

Don't spend too much time

Prototyping is all about speed; the longer you spend building your prototype, the more emotionally attached you can get with your idea, thus hampering your ability to objectively judge its merits.

Remember what you're testing for

All prototypes should have a central testing issue. Do not lose sight of that issue, but at the same time, do not get so bound to it so as to lose sight of other lessons you could learn from it.

Build with the user in mind

Test the prototype against your expected user behaviors and needs. Then, learn from the gaps in expectations and realities, and improve your ideas.

Stage 5: Test—Try Your Solutions Out

Test

Test: the fifth and final phase of the design thinking process, where you test solutions to derive a deep understanding of the product and its users.

Designers or evaluators rigorously test the complete product using the best solutions identified in the Prototype stage. This is the final stage of the five-stage model; however, in an iterative process such as design thinking, the results generated are often used to redefine one or more further problems. This increased level of understanding may help you investigate the conditions of use and how people think, behave and feel towards the product, and even lead you to loop back to a previous stage in the design thinking process. You can then proceed with further iterations and make alterations and refinements to rule out alternative solutions. The ultimate goal is to get as deep an understanding of the product and its users as possible.

5 Guidelines for Conducting a Test

1. Show, don't tell: let your users experience the prototype

Make sure to introduce yourself. Never, ever say you are the designer, even if you are. People will be less honest with feedback if they think you are the author and won't want to hurt your feelings. Explain how long the session should take, what your expectations are for them and what they are going to be doing. Always ask if they have any questions before starting.

Avoid over-explaining how your prototype works, or how it is supposed to solve your user's problems. Let the users' experience in using the prototype speak for itself, and observe their reactions.

2. Ask Participants to Talk Through Their Experience

When participants are exploring the prototype, ask them to tell you what they're thinking. Let them know that they should think out loud and speak what's on their minds during the entire test session. This doesn't come naturally to people, so you may have to prompt the participant during the test to remind them. In your intro to the test, make sure to let them know you are expecting this and give an example. You want them to let you know what they expect to happen when they select something or what they were expecting to see on the screen based on the title or location.

3. Observe Your Participants

Be a neutral observer. Observe how your participants use your prototype and resist the urge to correct them when they misinterpret how it's supposed to be used. Mistakes are valuable learning opportunities. Remember that you are testing the prototype, not the participant.

4. Ask Follow-Up Questions

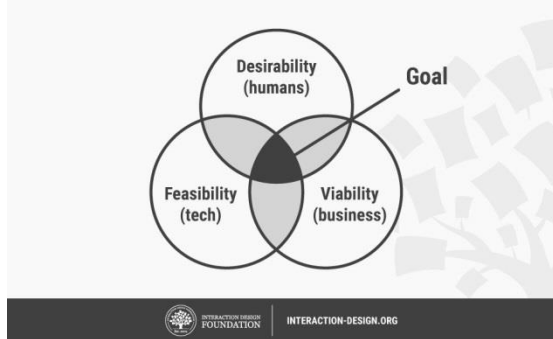
Always follow up with questions, even if you think you know what the participant means. Ask questions such as "What do you mean when you say ____?", "How did that make you feel?", "What did you expect

would happen?” and, most importantly, “Why?”.

5. Negative Feedback is Your Way to Learn and Improve

When you test your ideas and prototypes, remember that negative feedback is an important way to learn and improve. You might feel a sting in the moment when you hear a person complain about how difficult your prototype is to use, but try to get used to the idea that such feedback will help you in the long run. You will uncover problems that you and your team might not have even considered.

The End Goal: Desirable, Feasible and Viable Solutions



The design thinking process doesn't follow a fixed sequence of steps, but it has an ideal end point. The end goal of every design thinking project is a solution that is desirable, feasible and viable.

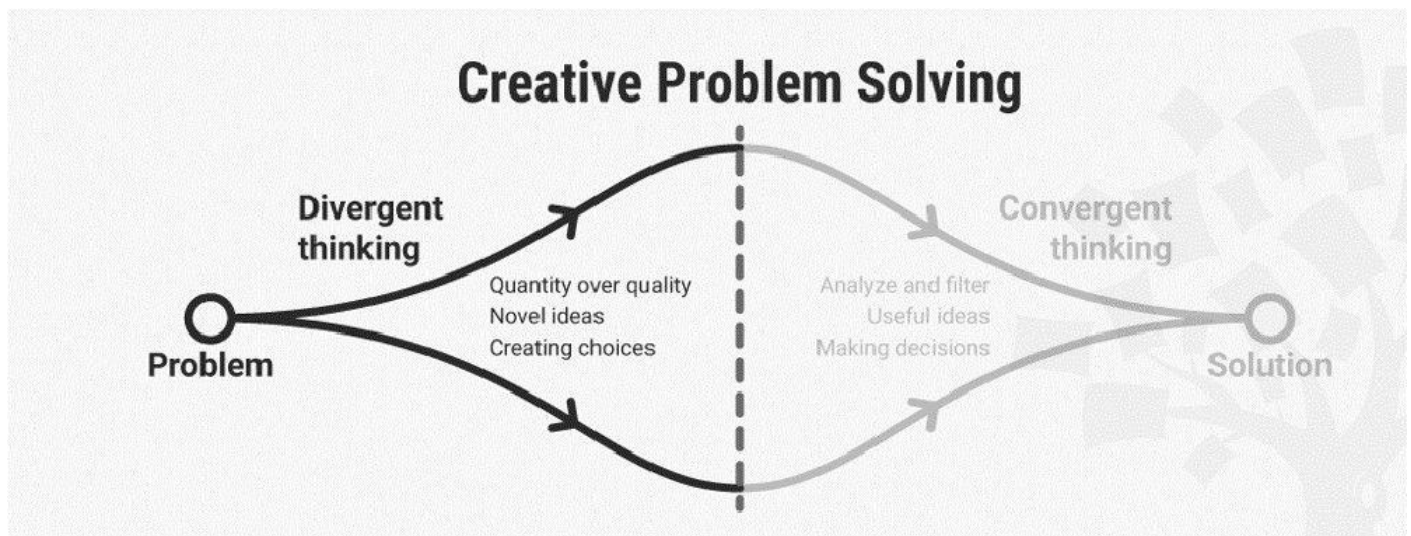
Desirability focuses on people. It's what puts the “human” in human-centered design. Your solution is desirable if it appeals to the needs, emotions and behaviors of the people you target.

Feasibility is about technology. Is your design solution technically possible or does it depend on a technology that's yet to be invented (or good enough for regular use)?

Viability is about whether your design solution works as a business. Is there an appropriate business model behind your solution, or would it collapse after a few years without investor or donor contributions? Design thinking is not about making a profit, but good design solutions should be self-sustaining. That way, you can continue to support and improve your solution way beyond the project deadline.

Overall, you should understand that these stages are different modes which contribute to the entire design project, rather than sequential steps.

DIVERGENT AND CONVERGENT THINKING



What is Divergent Thinking?

Divergent thinking is an ideation mode which designers use to widen their design space as they begin to search for potential solutions. They generate as many new ideas as they can using various methods (e.g., oxymorons) to explore possibilities, and then use convergent thinking to analyze these to isolate useful ideas.

Divergent Thinking Can Open up Endless Possibilities

The formula for creativity is structure plus diversity, and divergent thinking is how you stretch to explore a diverse range of possibilities for ideas that might lead to the best solution to your design problem. As a crucial component of the design thinking process, divergent thinking is valuable when there's no tried-and-tested solution readily available or adaptable. To find all the angles to a problem, gain the best insights and be truly innovative, you'll need to explore your design space exhaustively. Divergent thinking is horizontal thinking, and you typically do it early in the ideation stage of a project. A "less than" sign (<) is a handy way to symbolize divergent thinking – how vast arrays of ideas fan out laterally from one focal point: Design team members freely exercise their imaginations for the widest possible view of the problem and its relevant factors, and build on each other's ideas. Divergent thinking is characterized by:

Quantity over quality – Generate ideas without fear of judgement (critically evaluating them comes later).

Novel ideas – Use disruptive and lateral thinking to break away from linear thinking and strive for original, unique ideas.

Creating choices – The freedom to explore the design space helps you maximize your options, not only regarding potential solutions but also about how you understand the problem itself.

Divergent thinking is the first half of your ideation journey. It's vital to complement it with convergent thinking, which is when you think vertically and analyze your findings, get a far better understanding of the problem and filter your ideas as you work your way towards the best solution.

A Method to the "Madness" – Use Divergent Thinking with a Structure

Here are some great ways to help navigate the uncharted oceans of idea possibilities:

Bad Ideas – You deliberately think up ideas that seem ridiculous, but which can show you why they're bad and what might be good in them.

Oxymorons – You explore what happens when you negate or remove the most vital part of a product or concept to generate new ideas for that product/concept: e.g., a word processor without a cursor.

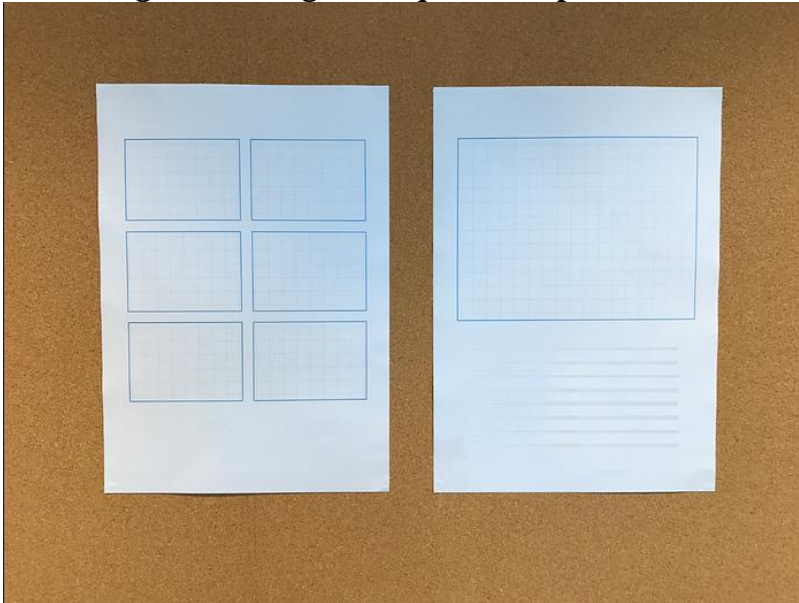
Random Metaphors – You pick something (an item, word, etc.) randomly and associate it with your project to find qualities they share, which you might then build into your design.

Brilliant Designer of Awful Things – When working to improve a problematic design, you look for the positive side effects of the problem and understand them fully. You can then ideate beyond merely fixing the design's apparent faults.

Arbitrary Constraints – The search for design ideas can sometimes mean you get lost in the sea of what-ifs. By putting restrictions on your idea—e.g., “users must be able to use the interface while bicycling”—you push yourself to find ideas that conform to that constraint.

Divergent thinking has been a core technique of good ideation in the design and creative industries. But there's still a lot of potential to expand these techniques further and add value to other disciplines such as sales and operations.

The divergent thinking technique — 6-Up to 1-UP used in SEEK company



It's a structured framework that goes along these lines:

Consider the challenge

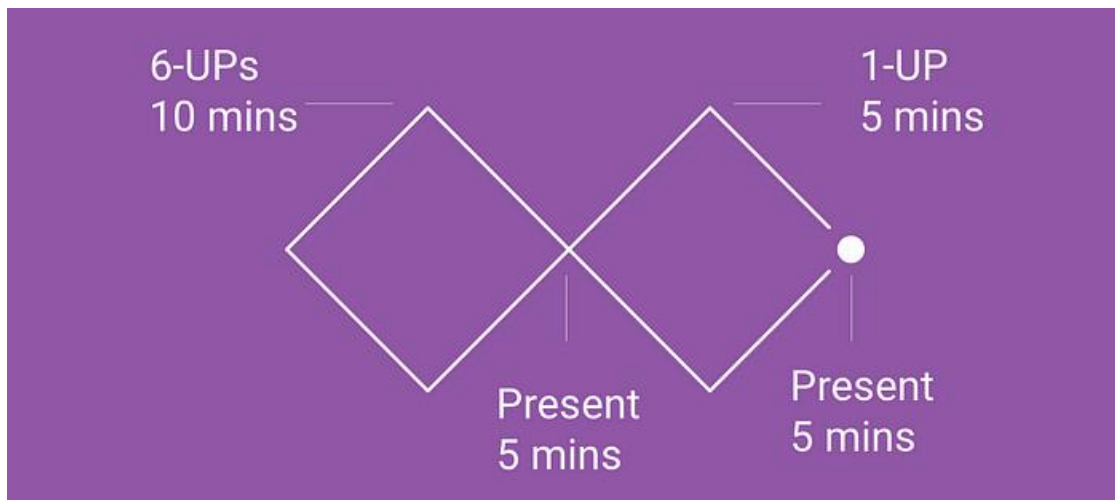
Diverge by spending 10 minutes drafting 6 different ideas (6-UP template)

Present each idea for a minute and seek feedback

Converge by developing the most promising idea further

Diverge again (1-UP template)

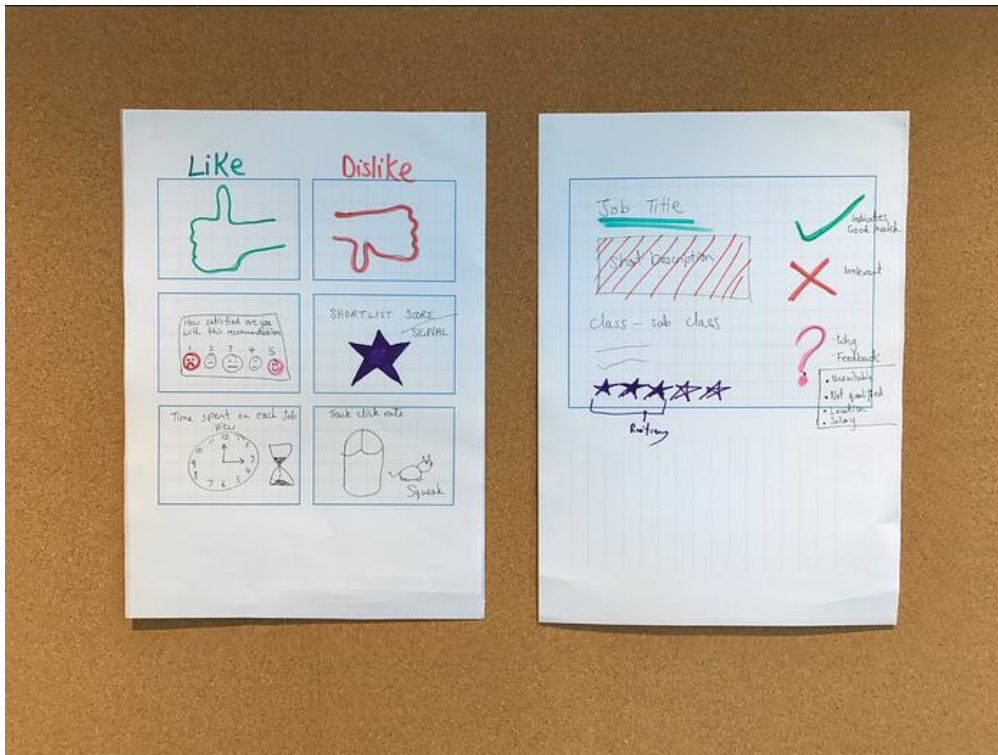
Converge by presenting, or consider further and seek out feedback.



At this point you will have very quickly:

grown your first idea into a number of ideas
received some early feedback and
increased your likelihood of ending up with a better solution.

The great thing about divergent thinking is that it's not about design, it's about ideation. So regardless of the discipline, generating ideas in this structured way can be useful for all sorts of things.



The great thing about divergent thinking is that it's not about design, it's about ideation. So regardless of the discipline, generating ideas in this structured way can be useful for all sorts of things.

While divergent thinking is not a design-only practice, it's commonly used in the design and creative industries. But there is the potential to use these techniques elsewhere.

How to Think and Work Divergently – 4 Ideation Methods

1. Oxymorons

With this method, you explore what happens when you remove what's most essential about a product or concept (e.g., a service) to generate new ideas for the product or concept.

Oxymorons is a divergent ideation method. With the Oxymorons method, you explore what happens when you remove what's most essential about a product or concept to generate new ideas for that product or concept. When you set up a constraint like this, you force yourself to think about new uses that you might not have thought about for the original product. The trick is to think about the product or idea you're working on, write down what its most essential characteristics are and take it from there. It will help you to know that the word oxymorons means rhetorical maneuvers that use seeming self-contradictions to reveal a paradox or illustrate rhetorical points. Oxymorons are also used in general as a contradiction in terms when you want to describe something which appears to be a contradiction, but is actually possible—for example, “living dead” or “bitter-sweet”.

This is What You Do

1. Choose a product or a service and figure out, “What's the essential characteristic?”

Think of a concrete feature, not something abstract

2. Imagine what the product would be like without that essential characteristic. Imagine, for example, if you had a word processor without a cursor, you couldn't choose where to write; or if you had a PowerPoint presentation, but you didn't have a projector to show your slides during your presentation.

3. Write down your ideas for what the product without the essential characteristic could be used for—or imagine other contexts in which it could be used. For example, a word processor without a cursor could be used in a creative writing exercise where you practice not editing what you have written. If the word processor places the cursor randomly, you might use it to make some form of abstract poetry. If you were not in the context of working, but wanted to force yourself to take time off, having the cursor disappear from your word processor would force you to work in a different manner or to do something other than working.

4. Then consider your original product again. Could you make any improvements or come up with any new features inspired by the ideas you generated in step 3? For example, think about ways to help users take breaks from writing, or ways to make it easier to create playful content that doesn't fit into the horizontal lines of a typical word processor.

Best Practice

You can use the method as an individual or in a group.

It's a good idea to decide beforehand how much time you want to spend on each part of the exercise, especially in a group setting.

Do a test run on your own and see how much time you spent if you're unsure on what's fitting for your project

2. Random Metaphor

This method pushes you to think about your project in new ways, so it's great when you want to think more broadly about the idea or product you're working on.

Random Metaphors is a divergent ideation method. It pushes you to think about your project in new ways, so it's a great method when you want to think more broadly about the idea or product you're working on.

1. Pick a random item in your surroundings or a random word in a dictionary. Write down as many attributes and associations as you can think of for the object you've picked. For example, if you pick a houseplant, you could write: "green, alive, growing, soil, needs to be taken care of, photosynthesis".

2. Pretend that a genius in your field told you that the thing you just picked is a good metaphor for an item/word/concept you're working on. Come up with an explanation for how the thing you picked is a good metaphor. For example, a plant is a great metaphor for a word processor because, they both grow over time (a word processor grows when you add new features), they both need to be taken care of (for a word processor you have to be sure to provide support for the users and fix bugs in the programming). As you can see from our examples, the metaphors don't have to be great. In fact, they probably won't be—since you picked a totally random item

3. Use the metaphors you came up with to improve or change your product. For example, come up with ways to ensure that new features in a word processor are introduced organically so they seem like a natural evolution to ensure that users won't be confused when they encounter something new

Best Practice

You can use the method as an individual or in a group.

It's a good idea to decide beforehand how much time you want to spend on each part of the exercise, especially in a group setting. Do a test run on your own and see how much time you spent if you're unsure on what's fitting for your project.

3. Brilliant Designer of Awful Things

You can use the method when you work to improve a design or product that has one or more known problems. The method helps you see the positive sides of the problem to ensure that you fully understand it

before you make changes to a design and to help you ideate about ways to improve the product beyond just fixing what is wrong with it.

Brilliant Designer of Awful Things is a divergent ideation method. You can use the method when you work to improve a design or product that has one or more known problems. The method helps you to see the positive sides of the problem to ensure that you fully understand it before you make changes to a design and to help you ideate about ways to improve the design or product beyond just fixing what is wrong with it.

Follow These 4 Steps

1. Write down the features which are problematic in your product or idea. For example, you've got a battery level indicator on a camera with just 2 levels: 50% and 100%.
2. Analyze why each of the features you wrote down is problematic. For example, it's problematic that the battery level indicator can show 50% but then nothing smaller before it runs out of power. The user has no warning of when she needs to charge it.
3. Imagine that each of the problematic features was designed by the most brilliant designer you know—someone who usually does things for a good reason. Then try to come up with reasons for why the feature was designed the way it was. For example, you could have: “Adding a third battery level indicator would take up space on the tiny screen; we would then have to create a bigger screen, and that would leave no space for the flash because it's a compact camera.” or “User research uncovered that users are more likely to charge the camera right away instead of waiting until the last minute and thus less likely to run out of power when they don't know what their battery level is below 50%.”

4. Consider the good reasons in step 3. Is there a grain of truth in some of the good reasons that you need to consider when you create the redesign? Do any of them spark related ideas for your redesign? For example, could you design something that nudges your users towards charging their camera? For instance, maybe you could design a camera stand that doubles as a charger.

Best Practice

You can use the method as an individual or in a group. It's a good idea to decide beforehand how much time you want to spend on each part of the exercise, especially in a group setting. Do a test run on your own and see how much time you spent if you're unsure on what's fitting for your project.

4. Arbitrary Constraints

This is a great method to inspire ideation and think outside of the box, because you push yourself to think about the idea or product you're working on in new ways when you place arbitrary constraints on your ideation.

Arbitrary Constraints is a great method to inspire ideation and think outside of the box, because you push yourself to think about the idea or product you're working on in new ways when you place arbitrary constraints on your ideation.

Here's What You Do:

1. Write down 10–20 arbitrary constraints on individual pieces of paper and place all the pieces face down. For example, here are some constraints for a mobile phone: “use while bicycling” or “to be used by a blind person”. You can also print out our list of arbitrary constraints, cut them out and place them face down.
2. Think about the idea or product you're working on and then pick a random piece of paper with a constraint on it. For example, a mobile phone... “that you can use while bicycling”.
3. Come up with as many ideas as possible for your product that works with the arbitrary constraint. For example, the user can wear a headset and call up people on their contact list via voice control. Keep working until you run out of ideas.

4. Look through your ideas and consider how each idea could make sense without the arbitrary constraint. For example, it would make sense to be able to call people on your contact list using a headset in many other situations than bicycling

Arbitrary Constraints

Here are some arbitrary constraints to help you get started:

We've also converted these arbitrary constraints into cards that you can cut out and use!

You'll find them at the end of this template.

Arbitrary Constraints

can be used while bicycling

can be used by a blind person

something we can create in 1 week

can be used by a 3-year-old

can be used no matter what your language is

can be used by multiple users at the same time

can be used underwater/while in water

something we can create without a budget

can be used in complete darkness

can be used with no hands

can be used with no internet connection

must have physical controls

Best Practice

You can use the method as an individual or in a group.

It's a good idea to decide beforehand how much time you want to spend on each part of the exercise, especially in a group setting.

Do a test run on your own and see how much time you spent if you're unsure on what's fitting for your project.

Three Phases and Modes of Thinking in Ideation

Let's create an overview of the various modes of thinking in ideation and in creativity. Even though we break them up here to grasp them, it's important to note that these thinking modes work together. In fact, they are deeply dependent on each other.

You use the divergent thinking mode in the early stages of your creative ideation sessions. At this stage, it's important that you get lots of new ideas. You broaden up your design space, and you should envision your possibilities as endless. Your thinking will often start from a single point of focus, and then you'll expand further and further as you create new and unique ideas. This divergent phase is a time for disruptive and lateral thinking. Divergent thinking is a generative cognitive activity, where the quantity of ideas is more important than the quality.

B) Emergent Thinking

Emergent thinking is an in-between thinking style which leverages ideas generated in the early divergent stage as stimuli to allow the generation of additional ideas. It's a phase where you and the rest of the ideation group build upon initial ideas to make new connections and combine and spark new ideas. The early, more chaotic divergent sessions should provide a wide enough range of stimuli to make this possible—and, again, disruptive lateral thinking should be utilized.

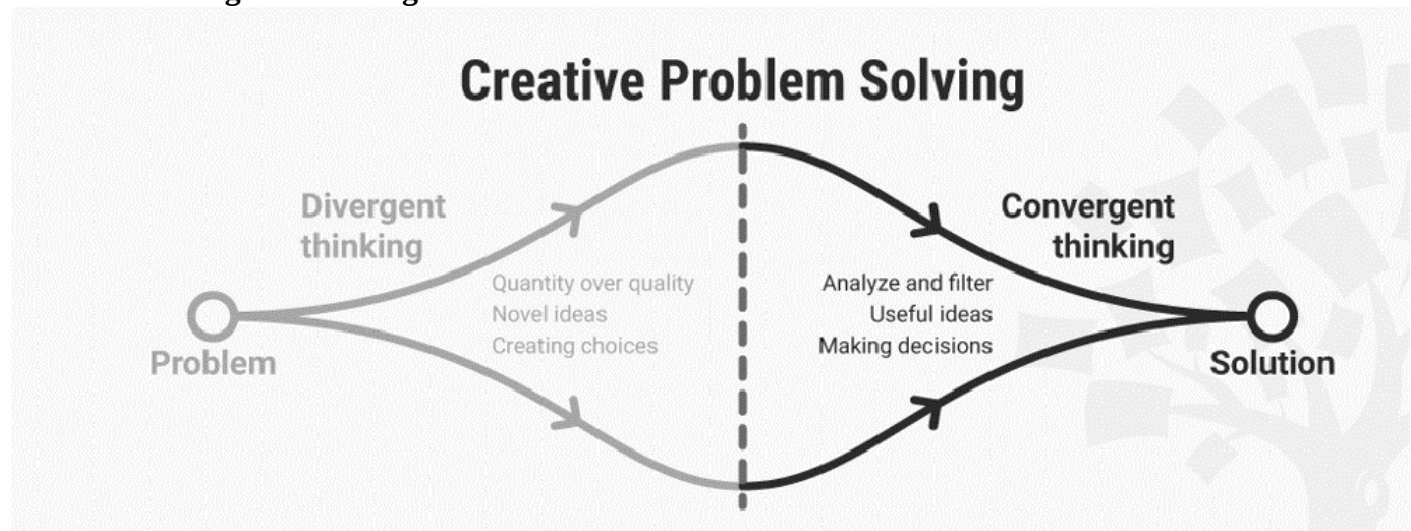
C) Convergent Thinking

Convergent thinking occurs towards the end of ideation sessions and closes off these sessions through

allowing your team to sift through ideas, group them into themes, pick out common threads and ultimately decide upon winners and losers. In this sense, it's good to remember that ideation is not meant to be a wild frenzy of chaos. Impractical or impossible solutions do need to be weeded out, and the overly ambitious ones need to be shelved for later consideration in certain cases. This convergent phase is therefore where you make decisions through the lenses of desirability, viability and feasibility. Your creative process shouldn't end with these decisions, though—this is merely the first time you will prune and prepare your best ideas ahead of the rest of your creative process: Let's say you're creating a new app. In this case, you would of course want to prototype and test it and not leave it at this ideation stage.

Convergent thinking relies on a more vertical, linear and analytical style of thinking, though using it means you all still need to maintain a good degree of creativity to ensure ideas with merit are not rejected merely because they do not follow logical norms.

What is Convergent Thinking?



Convergent thinking is an ideation mode which designers use to analyze, filter, evaluate, clarify and modify ideas they have generated in divergent thinking. They use analytical, vertical and linear thinking to find novel and useful ideas, understand the design space possibilities and get closer to potential solutions.

Convergent Thinking – the Other Side of the Ideation “Coin”

After design teams generate as many ideas as possible in the divergent thinking part of ideation sessions, convergent thinking helps them systematically see whether their ideas might work as real-world solutions. The structure is to:

Sift through ideas.

Group them into themes.

Find common threads.

Decide on winners and losers.

Convergent thinking helps tighten your focus when evaluating each idea. For example, if your design problem concerns users with errands, one idea might be an app for users to control their cars remotely to

send/collect goods. You'd then examine it through three lenses:

Desirability – “Would users want this?” (Or would they fear accidents, hacking, theft, etc.?)

Viability – “Could a brand mass-produce and support it?” (Or would it be unsustainable/too expensive?)

Feasibility – “Is it doable?” (Or would security, sensory and emergency-backup features take years/decades to perfect?)

Then, considering state-of-the-art technology and other factors, you might abandon this idea as impracticable or shelve it for future consideration.

See Ideas in a New Light with Convergent Thinking

Convergent thinking isn't a clinical process that automatically results in optimal solutions. Rather, you need a careful, creative mindset to:

Look past logical norms (which we use in everyday critical thinking);

See how an idea stands in relation to the problem; and

Understand the reality/dimensions of that problem.

As you work more insightfully, you can begin to understand the idea in the context of what's going on in the problem domain. And only with an accurate understanding of the problem can you determine the best criteria to judge an idea with. Otherwise, it's easy to overlook the problem domain's complexity and apply just your existing knowledge (e.g., “It looks like something that a mega menu could help with.”). Some dimensions of it may be unlike anything you'll recognize. By studying problems and ideas on their own terms, though, you can avoid misidentifying them with assumptions.

Thinking convergently helps overcome many obstacles, even if alternative ideas can also cause problems. Sometimes, a clearer understanding might show you the best solution straightaway. Or you might use that understanding to generate new ideas and newer understandings. Whether you're fine-tuning novel ideas through thought-provokingly fresh lenses or suddenly finding yourself inspired to work on a prototype to test, convergent thinking helps advance your creative process. That's why it's vital in a design process such as design thinking.

How to Use Convergent Thinking to Find Novel and Useful Ideas

Some techniques to help you focus creatively are:

Externalization – You sketch ideas to make all the tacit knowledge and underlying assumptions about your project visible and then identify that knowledge (including its limits) to help understand the problem domain.

Three-Way Comparisons – You compare three similar products to see how each differs from the other two.

What are 3-Way Comparisons?

Three-way comparisons are a convergent ideation method designers use to map their assumptions and tacit knowledge about familiar domains to new domains. They compare three related items in all possible combinations to uncover hidden attributes, understand the problem domain better and find insights to use innovatively.

Compare Three Things – Find More in Each

A vital skill underpinning a designer's creativity is to bring tacit knowledge out into the open where it can be analyzed and insights shared. Sometimes, we have deep-held assumptions about designs, concepts, etc. These can range from an item's smallest qualities up to critical factors of the mental model users adopt when encountering problems in context. If we take things for granted, we might overlook crucial points. That's why it's essential to isolate errors in how we envision design-related matters, and hopefully pinpoint previously unheard-of ways to access users.

Whenever we compare similar things, we can clearly map what makes them distinct as we systematically examine each's qualities. The simplest way is to compare two items; by asking what's similar and different, you can quickly compile a list. So, by identifying and discussing the distinctions between (e.g.) a menu on a webpage versus on a phone, you can make sense of the various characteristics of each.

However, a three-way comparison can help you uncover far more. This approach comes from the repertory grid technique used to map personality traits. It's especially useful for exposing and expressing qualities which might otherwise lie buried under assumptions and bias. For example, you might compare a dropdown menu with a cascading menu with a pie menu. By starting with "How is A like B but different from C?" you proceed through different orders of comparison (e.g., "How is C like A but different from B?"). Consequently, you'll force yourself to stretch beyond those obvious standard differences and engage the subjects more open-mindedly. In our video example, distinctions include:

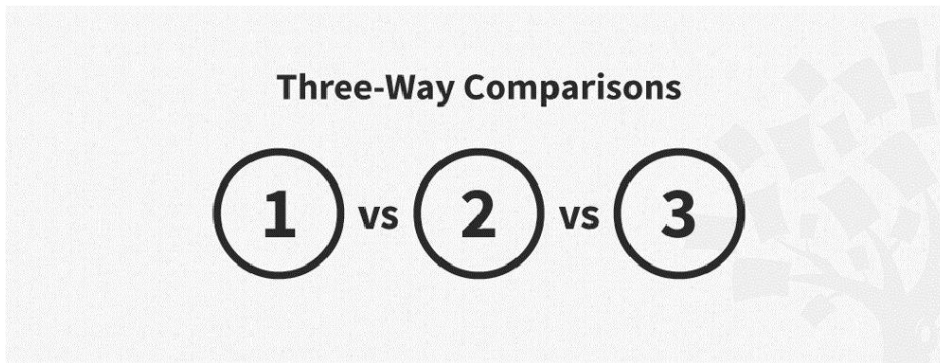
Grapes are smaller – one serving contains more of them.

Apples are contiguous, whereas oranges are segmented.

Oranges are the "messiest" to eat.

Grapes are to wine as apples are to cider – the priority you give helps define how you see the world.

By making these distinctions, you expose the categories you've involved. Also, when you list these inherent qualities, categories, etc., it's far easier to spot new differences more easily and (potentially) new contexts of use.



How to Use 3-Way Comparisons

1. Pick three designs/products in the domain you're interested in. (One of these can be your own if you work in that domain.) They should be similar, so you/your users must put effort into considering differences between them. If, for example, you're designing a new user interface menu and you want to understand the domain better, you can choose three existing menu types for comparison: dropdown versus cascading versus pie.

2. Compare each design to the other two. Here's where you identify your own assumptions. Write down how each differs from the others. For example, from comparing how the dropdown differs from the cascading and pie menus, you notice a dropdown menu is horizontal and simpler.

Keep going as long as possible. If stuck, try considering a different dimension in which to compare the menus.

3. To make comparisons involving your users: Sitting down with one user at a time, show them the different designs and ask them to compare these: e.g., "Please list as many ways as you can think of how a dropdown menu is different from a cascading menu and a pie menu."

Write down what each user lists as different. Users will likely soon get stuck, but encourage them to persevere by assuring them that no comparisons are silly. You can also encourage them to compare the designs in other dimensions which they haven't considered yet. (Note: As users can find this hard, it's best to keep adding users until no new attributes are mentioned.)

4. Consider the lists of attributes in relation to the domain you want to design for: After making comparisons, you'll have three lists of attributes (one for each design). If you've involved users, you should combine the users' lists, so you have a list for each design that summarizes what every user said.

More Specific and More General – You place some temporary constraints on your project to get a more specific overview to help you generate ideas, and you stand back to get a more abstract overview.

Embrace Opposites – You find overlaps between different categories or opposites to spot new design opportunities.

Embrace opposites is a convergent ideation method which designers use to explore their design space by finding overlaps between different categories or opposites. When they chart and compare two apparent opposites, they might find features that are common to both—and ones that are not—and spot new design opportunities.

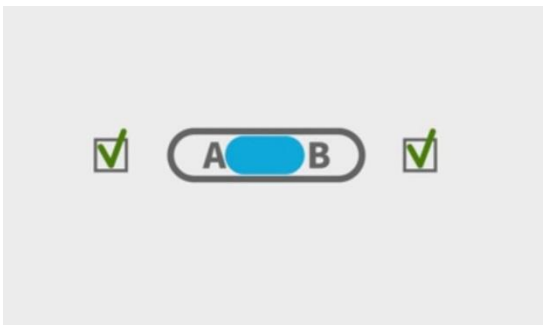
Appearances can Deceive – Go Behind the Scenes with Embrace Opposites

In our complex world, it's simpler to differentiate things by thinking in terms of dichotomies or opposites (e.g., books and websites) even when they have overlapping attributes. Distinguishing things this way helps us make quick decisions: (e.g.) “good” or “bad” as absolute values, without considering the many degrees in between that describe something's/someone's qualities. However, reality is usually too complicated to categorize with “either/or” labels. Often, things that seem totally opposed (e.g., political parties, personalities) share characteristics. For example, what does an “introvert” look like? Or an “extrovert”? Can someone be both?

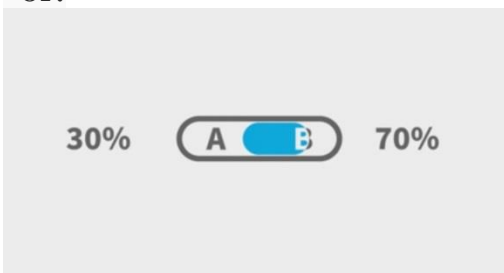
In ideation, you can embrace opposites to see if you can enrich a problem and focus on designing innovative features. For example, consider a simple switch:



If A is “Off” and B is “On”, these are categorical distinctions. However, if A and B were other items that were opposed or distinct (e.g., menus and radio buttons), you might see them in dimensional terms, instead, and ask if they share features. Also, you might be able to design for a combination of these, perhaps with more of one than the other:



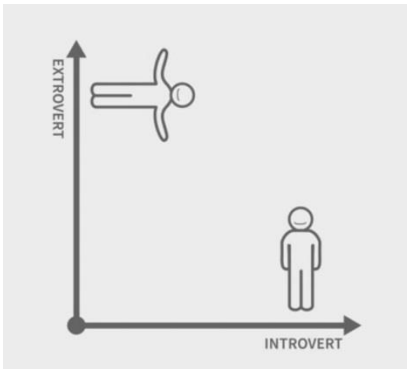
or:



So, you can get a deeper understanding of a design problem and the elements you're working with if you analyze the categories and dichotomies you perceive. Dimensions tend to be richer than categories, but trickier to work with. To envision this, let's re-approach our extrovert-introvert dichotomy. We might flip it into a graph, so:



becomes:



This makes it easier to look for common elements and neutral ones if we divide our graph into 4 squares, where:

Top left = More Extroverted	Top right = Design Possibilities!
Bottom left = Neutral	Bottom right = More Introverted

So, you might find a dimension to manipulate in your own project. It might be (e.g.) a menu (A) that pulls down and includes radio buttons (B) – anywhere where elements of both apparent opposites work simultaneously. At least, you can confirm when “opposites” are indeed distinct.

How to Embrace Opposites

Try these steps to identify design possibilities:

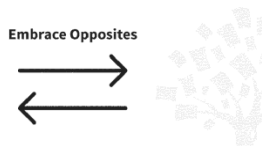
Create an overview of the different categories or opposites you have in a current design problem. (E.g.) You might say that the desktop version of an application and the mobile version of the application belong in two different categories: desktop and mobile.

Dissolve the categories and ask yourself “In what ways can my application be both desktop and mobile?”. On the surface, it can’t (users either use their phones or desktops to view your application). But are there any situations where it can? (E.g.) a tablet is in some ways in between the two. It has a big screen like a desktop, but a smartphone’s touchscreen interactivity.

List all the overlaps you find.

Go through your list and consider how big the overlap for each item is, whether it’s mostly one or the other (remember our x-and-y-axis graph above). Now place the items on your list there. E.g., a tablet is probably midway between a desktop and a smartphone, but a laptop with a touchscreen is more like a desktop computer than a smartphone.

Consider which consequences the overlaps have for your design. Should you use the same interaction principles for all of these devices? If not, how many versions do you need? If your application is specialized, maybe the mobile with a touchscreen is the perfect device for it and you should forget about all the other versions.



Make Things Opposite

Embrace opposites is also helpful if you disagree with the design goal:

Reverse your problem statement. E.g., you wanted to help people feel more inclined to use public transportation than their cars. Now, imagine you wanted to make people more inclined to use their cars. What would that take? How can you try to make people spend even more time in their cars? How can you make it even nicer to drive the car and less attractive to use public transportation? E.g., you could:

Simply decrease the number of bus routes in the area;

Have it so there's only one bus in the morning, one in the evening;

Make bus tickets more expensive;

Lengthen the routes;

Have the driver play really loud techno music on routes which elders often use.

Now, turn those insights around. E.g., you could research to see what areas most people are taking busses in. Then, you could make more busses available more frequently in those areas and offer fewer busses where there are rarely any passengers.

Overall, remember that determining the opposite of something can be complicated. However, the effort can pay big dividends.

Multiple Classifications – You compare aspects of items using a matrix to widen your inspiration, spot market gaps, analyze trends and rules and examine related qualities.

What are Multiple Classifications?

Multiple classifications is a convergent ideation method where designers explore the design space to find opportunities for new products. Using matrices, they compare aspects of items to widen their inspiration, spot market gaps, find and analyze trends and rules, and see if related qualities hint at inventive designs.

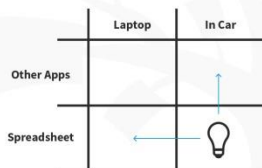
Explore More Possibilities with Multiple Classifications

A single-classification system (or taxonomy) is ideal for organizing items in libraries, computer folders, etc. Also, it's easier to list similarities between (e.g.) two types of fish than between fish and birds. The downside to categorizing items this way, though, is the need to put things with similar features in one place. That can obscure other similarities and block insights.

Multiple classifications can help you find and understand how items are similar while you analyze their differences. Consider two circles and two squares – one red, one yellow of each. How would you group them: by shape or color? Using a taxonomy, you'd describe one common attribute at a time. However, with multiple classifications, you can describe both simultaneously. And, in ideation, you can explore and map your design space far more extensively and tap powerful market opportunities.

Enter a Matrix!

You can use several different, but related, methods:



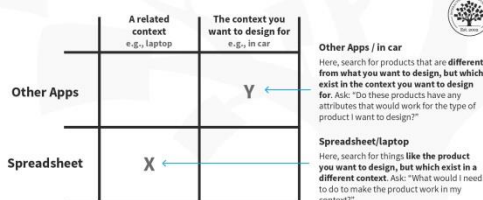
Interaction Design Foundation
interaction-design.org

1. Spread your search for inspiration – To develop a new design concept: e.g., productivity apps for autonomous (self-driving) car users:

a. Draw a 2-by-2 matrix – on one axis, write the context you want to design for (here, “car”) and another context that’s different but still relevant.

b. On the other axis, write the type of product you want to design (e.g., “productivity apps/spreadsheets”) and then one that contains all other kinds of products.

This matrix gives you 4 categories. You’ll want to analyze 2 (“X” and “Y”):



Interaction Design Foundation
interaction-design.org

2. Identify Gaps – In our example, you could notice that many apps didn’t include spreadsheets. So, you could see if an app that uses spreadsheets might help autonomous car users.

a. Using your matrix, find all the systems (e.g., books) discussing the area you want to research (so, spreadsheet/laptop and other apps/in car).

b. Complete your matrix with the information you’ve found. E.g., you should have items to insert in 3 spaces in the matrix’s category pairs, but now you’ll likely notice a gap (e.g., in the bottom-right corner). This could be a market opportunity.

3. Analyze and Discover Trends and Rules – If you’ve found a pattern in your design space, it’ll be far easier

to design relevant products if you find and understand the rules and trends in that space.



		Are Prices Shown?	
		No	Yes
Brand	Volume	?	B&Q New Look
	Exclusive	Hermes Mouawad	?

Interaction Design Foundation
interaction-design.org

a. Draw a matrix with the categories you expect to find and start to research.

b. E.g., you notice a trend that exclusive brands' websites don't advertise prices, but volume brands' do. So, draw a matrix of whether different website types advertise prices. Research websites and put your findings in the matrix to see if you're right.

In our example, the pattern is predictable. However, if you find that the pattern looks different from what you expect, ask yourself why. Can you deduce any general rules? What does that mean for your design? You may have discovered a space where no other products exist.

4. Uncover Abstractions – To see if a general rule you've spotted might apply for all items (e.g., devices) of that type and, if so, why.

On a 2-by-2 matrix (or larger) mark the categories you'll compare and explore.

E.g. (below), imagine you've spotted a general rule about fitness apps for running and cycling, since specialist devices (e.g., Fitbits) in one column share a property (they show the heart rate) and non-specialist devices such as smartphones share another (they don't show it). Ask yourself if this generalization holds true for all phones versus specialist devices.

Search for the reason (e.g., because phones lack heart-rate sensors). So, now, having noticed a relationship between types of designs (e.g., devices suitable for a fitness app), you can explore the potential for what one design can do.

This abstracted knowledge is your understanding of why the relationship exists. Then, you can reapply it to see if you can design things on your matrix's left-hand side (so, generalist devices such as smartphones) that have similar features/capabilities to those on the right-hand side. For example, maybe a phone could track someone's heart rate in another way (e.g., through its microphone or other sensors).



		Non-specialized devices e.g., smartphone	Specialist devices e.g., Fitbit
	Running	do not show heart rate	show heart rate
	Cycling	do not show heart rate	show heart rate

Interaction Design Foundation
interaction-design.org

Overall, good judgment is essential. Sometimes, there'll be a valid reason a product doesn't exist (e.g., it would be hazardous). Sometimes, though, you might find you've stumbled on a lucrative market gap.

Teasing Apart, Piecing Together (TAPT) – You break down an experience into its elements, reconstruct it with a better understanding of what's involved and re-imagine it in a new context.

BRAINSTORMING

Brainstorming

Brainstorming is a method design teams use to generate ideas to solve clearly defined design problems. In controlled conditions and a free-thinking environment, teams approach a problem by such means as “How Might We” questions. They produce a vast array of ideas and draw links between them to find potential solutions.

How To Use Brainstorming Best

Brainstorming is part of design thinking. You use it in the ideation phase. It's extremely popular for design teams because they can expand in all directions. Although teams have rules and a facilitator to keep them on track, they are free to use out-of-the-box and lateral thinking to seek the most effective solutions to any design problem. By brainstorming, they can take a vast number of approaches—the more, the better—instead of just exploring conventional means and running into the associated obstacles. When teams work in a judgment-free atmosphere to find the real dimensions of a problem, they're more likely to produce rough answers which they'll refine into possible solutions later. Marketing CEO Alex Osborn, brainstorming's “inventor”, captured the refined elements of creative problem-solving in his 1953 book, *Applied Imagination*. In brainstorming, we aim squarely at a design problem and produce an arsenal of potential solutions. By not only harvesting our own ideas but also considering and building on colleagues', we cover the problem from every angle imaginable.

Everyone in a design team should have a clear definition of the target problem. They typically gather for a brainstorming session in a room with a large board/wall for pictures/Post-Its. A good mix of participants will expand the experience pool and therefore broaden the idea space.



Rules for Brainstorming:

Brainstorming may seem to lack constraints, but everyone must observe eight house rules and have someone acting as facilitator.

1. Set a time limit

d-school emphasises that the facilitator has to be intentional about setting aside a period when your team will be in “brainstorm mode”. In this time frame, it's the sole goal to come up with as many ideas as possible,

and during this period judgements of those ideas are prohibited. Typically, a Brainstorm will take around 15-60 minutes. It can be shorter or longer, depending on the difficulty of the problem and the motivation and experience of the group.

Best practice tip:

“Invest energy into a short period of time, such as 15 or 30 minutes of high engagement. Get in front of a whiteboard or around a table, but take an active posture of standing or sitting upright. Get close together.”
– dschool, Bootcamp Bootleg

2. Start with a problem statement, Point of View, How Might We questions, a plan or a goal – and stay focused on the topic

Alex Osborn, the father of the Brainstorming technique, emphasises that brainstorming sessions should always address a specific question or problem statement (also called a Point of View) as sessions addressing multiple questions are inefficient. Begin with a good problem statement or question. Design Thinkers and other ideation specialists have further developed this approach into the art of framing problem statements via methods. Specifically, these are “How Might We” questions.

“How Might We” Questions

Once you’ve defined your design challenge in a Point Of View (POV) statement, you can start to use “How Might We” (HMW) questions to reframe your POV and open up the design challenge to look for solutions. Instead of saying “We need to design X or Y”, for example, you can add “How Might We” to the beginning and ask “How might we design X or Y?”.

Why and When to Ask “How Might We”

The “How Might We” (HMW) technique maintains a level of ambiguity on purpose because its job is to open up the exploration space to a range of possibilities. It’s a rewording of the core user need which you uncovered through deep interrogation of the problem in the Empathize phase, and synthesized in the Define phase. Let’s break the question down into its component parts now to see how it achieves this task:

“How”

“How” suggests that we do not yet have the answer. “How” helps us set aside prescriptive briefs and explore a variety of endeavors, instead of merely executing on what we “think” the solution might be.

“Might”

“Might” emphasizes that our responses are possible solutions, not the only solution. “Might” also allows for exploration of multiple possible solutions, and means we won’t settle for the first idea that comes to mind.

“We”

“We” immediately brings in the element of collaboration. “We” reminds us that the idea for the optimal solution will most likely come from collective and collaborative teamwork.

“How Might We” (HMW) questions serve as a transition step between the Define and Ideate phases of design thinking. They help you identify topics that represent subsets of your POV before you move on to the mass generation of ideas in the Ideate phase. In this way, HMW questions provide the best way to open up brainstorm and other ideation sessions where you can explore ideas to solve your design challenge in an innovative and user-centered way.

The HMW method is cleverly constructed to open up the field for new ideas, admit that we do not currently know the answer to the design challenge and encourage a collaborative approach to design solutions. Your

HMW questions should be broad enough to generate a wide range of solutions during ideation sessions, yet narrow enough that specific solutions can be created for them.

For example, if your POV is: “Teenage girls need to eat nutritious food to thrive and grow in a healthy way.”—the HMW question may go as follows:

- How might we make healthy eating appealing to young females?
- How might we inspire teenage girls towards healthier eating options?
- How might we make healthy eating something which teenage girls aspire towards?
- How might we make nutritious food more affordable?

These are simple examples, all with their own subtle nuances that may slightly influence different approaches and techniques in the Ideate phase. Regardless, your HMW questions will ensure that your upcoming ideation and design activities are informed with one of more HMW questions that can spark the imagination of you and your team and align well with the core insights and user needs you’ve uncovered

Best Practice Guide to Asking “How Might We”

01: Ensure your POV hits the sweet spot: Without a statement of a clear vision or goal in a well-framed Point Of View, “How Might We” questions are obviously meaningless. The technique requires a well-framed objective (a POV in other words) which is neither too narrow nor too broad.

02: Always begin with your Point of View (POV): HMW questions should always relate to your POV. Rephrase and reframe your POV as several questions by adding “How Might We” at the beginning.

03: Break up your POV with multiple HMWs: You can digest and ideate on your design challenge further when you segment it up into smaller actionable and meaningful questions. Five to ten HMW questions for one POV is a good starting point.

04: HMW questions come before solutions: It’s often helpful (and logical) to brainstorm the HMW questions before the solutions brainstorm.

05: Check the breadth of your HMWs: Look at your HMW questions and ask yourself if they allow for a variety of solutions. If they don’t, broaden them. Your HMW questions should generate a number of possible answers and become a launch pad for your ideation sessions.

06: Ensure your HMWs are focused: On the other hand, if your HMW questions are too broad, you may need to narrow them down. You should aim for a scope that’s narrow enough to direct the start of your ideation sessions, yet broad enough to give you room to explore wild ideas.

3. Defer judgement or criticism, including non-verbal

The facilitator should always set a positive, unthreatening tone and tell participants to reserve criticism for a later critical stage in the ideation process. A Brainstorming session is not the time and the place to evaluate ideas, and you should avoid executional details. It’s crucial that participants are feeling confident by being in a safe environment so they have no fear of being judged by others when they put forward wild ideas. You should create equal opportunities for all participants. The best ideas often come from practitioners, students, and people who dare to think differently – and not necessarily only from the highly skilled and experienced managers.

Best practice tip: The facilitator should suspend judgement. This way, participants will feel free to generate unusual ideas. As a facilitator, you will find that it can even be a great idea to prohibit the word “no”. You’ll be surprised to see how effective this tool is and how it helps open up minds and creates a collaborative, curious, and friendly ideation environment. And you’ll find that it’s pretty fun too when team members have to initially say “yes” to even the weirdest ideas. Giving out half-thought-out ideas or strange suggestions is normally not socially accepted, is socially frowned on and leads to people holding back in normal situations.

Brainstorming deliberately gives permission to be “stupid” and “child-like”.

4. Encourage weird, wacky, and wild ideas

These new ways of thinking might give you better solutions.

"It is easier to tone down a wild idea than to think up a new one."

– Alex Osborn, the father of the Brainstorming technique

“Wild ideas can often give rise to creative leaps. In thinking about ideas that are wacky or out there we tend to think about what we really want without the constraints of technology or materials. We can then take those magical possibilities and perhaps invent new technologies to deliver them.”

– OpenIDEO, 7 Tips on Better Brainstorming

5. Aim for quantity

Aim for as many new ideas as possible. The assumption is that the greater the number of ideas you generate, the bigger your chance is of producing a radical and effective solution. Brainstorming celebrates the maxim “quantity breeds quality”.

6. Build on each others' ideas

As suggested by the slogan “1+1=3”, Brainstorming stimulates the building of ideas by a process of association. Embrace the most out-of-the-box notions and build, build, build. Be positive and build on the ideas of others. Brainstorming works well when participants use each other's ideas to trigger their own thinking. Our minds are highly associative. One thought easily triggers another. When we use the thoughts of others, then these will stop us getting trapped by our own thinking structures.

Best practice tip: Try to say and encourage others to say “and” instead of “but”. It takes practice, but this little trick works surprisingly well.

7. Be visual

Best practice tip: At IDEO, they encourage you to use coloured markers to write on Post-its and put them on the wall—or sketch your idea. Nothing gets an idea across faster than drawing it. It doesn't matter how terrible of a sketcher you are! It's all about the idea behind your sketch. And d.school has two great ideas for you:

“There are at least two ways to capture the ideas of a brainstorming:

1. Scribe: the scribe legibly and visually captures on the board ideas that team members call out. It is very important to capture every idea, regardless of your own feelings about each idea.

2. All-in: Each person will write down each of his or her ideas as they come, and verbally share it with the group. It is great to do this with post-it notes, so you can write your idea and then stick it on the board.”

– dschool, Bootcamp Bootleg

8. One conversation at a time

Listen to each other and elaborate on each other's ideas. Don't get obsessed with your own ideas. You're here to ideate together.



To capture everyone's ideas in a brainstorming session, someone must play "scribe" and mark every idea on the board. Alternatively, write down your own ideas as they come, and share these with the group. Often, design problems demand mixed tactics: brainstorming and its sibling approaches – braindumping (for individuals), and brainwriting and brainwalking (for group-and-individual mixes).

Risks of Brainstorming:

Brainstorming involves harnessing synergy – we leverage our collective thinking towards a variety of potential solutions. However, it's challenging to have boundless freedom. In groups, introverts may stay quiet while extroverts dominate. Whoever's leading the session must "police" the team to ensure a healthy, solution-focused atmosphere where even the shiest participants will speak up. A warm-up activity can cure brainstorming "constipation" – e.g., ask participants to list ways the world would be different if metal were like rubber.

Another risk is to let the team stray off topic and/or address other problems. As we may use brainstorming in any part of our design process—including areas related to a project's main scope—it's vital that participants stick to the problem relevant to that part (what Osborn called the "Point of View"). Similarly, by framing problems with "How Might We" questions, we remember brainstorming is organic and free of boundaries. Overall, your team should stay fluid in the search for ways you might resolve an issue – not chase a "holy grail" solution someone has developed elsewhere. The idea is to mine idea "ore" and refine "golden" solutions from it later.

Should you Forget Brainstorming?

Brainstorm, as great as it can be, has some obvious weaknesses, most of them human factors. Whenever a group of people gets together to work collectively on anything, we inevitably find some members dominating over others. Introverts take a back seat and prefer working away in a quieter self-reflective manner, while extroverts, especially egoistical ones, put a stake in the ground and try to own the show.

Here's the problem: Most pieces of research on brainstorming are rigorous but irrelevant to the challenge of managing creative work. For starters, comparing whether creativity happens best in groups or alone is pretty silly when you look at how creative work is actually done. At creative companies like IDEO, people switch

between both modes so seamlessly that it is hard to notice where individual work ends and group work starts.

Should Your Team Brainstorm as a Group or as Individuals?

Best practice: Switching between the two modes of individual and collective ideation sessions can be seamless—and highly productive. Alex Osborn's 1950s classic *Applied Imagination* gave advice that is still relevant: Creativity comes from a blend of individual and collective ideation.

Siblings of Brainstorming

Brainstorming (group sessions) has three siblings which you should get to know: Braindumping (individual sessions), Brainwriting (a mix of individual and group sessions) and Brainwalking (another mix of individual and group sessions).

It's often a good idea to do individual ideation sessions like braindumping, brainwriting and brainwalking before and after brainstorming group sessions.

Braindump – Individual Brainstorm

Braindump as a means to free up mental energy and allow freethinking. Holding onto your own thoughts, unfinished tasks, or unexplored ideas creates mental blockages and prevents freethinking. Furthermore, Braindump is an amazing technique to help quiet employees get a voice.

Should Your Team Brainstorm as a Group or as Individuals?

Best practice:

Switching between the two modes of individual and collective ideation sessions can be seamless—and highly productive. Creativity comes from a blend of individual and collective ideation

Best practice: How

If you were the facilitator, you'll brief ideation participants upfront on the problem statement, goals and important insights from previous research and findings.

Then ask all participants to write down their ideas as they come.

It's important that each participant does this individually – and silently.

Provide participants with sheets of paper, idea cards or traditional Post-it notes. Sticky notes are great, because they allow people to write their ideas down individually – one idea per note.

Give participants between 3 and 10 minutes to get ideas they have been thinking of off their chests.

After reaching the time limit of approximately 3-10 minutes, each participant will say a few words about his or her ideas and stick them on a board or wall. You should avoid initial discussions about notes when team members are presenting them. Ideas that come out of early braindump sessions should be shared verbally with the entire team in order to spark new streams of thinking or combinations of ideas.

While sticking the ideas up and presenting them, the group will also group duplicates together.

When all team members have presented their ideas, you can select the best ideas, which you

can continue to build and elaborate on in other ideation sessions. There are various methods you can use such as “Post-it Voting”, “Four Categories”, “Bingo Selection”, “Six Thinking Hats”, and “Now Wow How Matrix”.



Sticky notes are great for braindump sessions, because they allow people to write their ideas down individually – one idea per note. After reaching the time limit of approximately 3-10 minutes, each participant will say a few words about his or her ideas and stick them on a board or wall.

Brainwriting



Brainwriting is a technique where participants write ideas onto cards and then pass their idea cards on to the next person, moving those cards around the group in a circle as participants build on the ideas of others. Participants perform this technique in complete silence—and they are forced to build on, instead of criticise, other participants' ideas. The cycle can be repeated multiple times and can be applied to chunks of the problem being addressed, depending on the need. The beauty of brainwriting is that it levels the playing field immediately, and it removes many of the obstacles of group brainstorming. With traditional verbal brainstorming, the number of ideas which can be expressed at once is limited, and the time it takes to get through a number of ideas is much longer, which results in many participants forgetting or becoming confused while others shout out ideas. This is especially so for those who are shy or introverted or who may be at a disadvantage due to being less senior or unfamiliar with the specialisations being discussed.

Brainwriting is an excellent starting point for ideation sessions, and can serve as a means to maximise the initial braindump, or as a way to refocus if other ideation methods go haywire. Before the chaos of group ideation muddles people's thinking, help them get their initial thoughts out in the open with an introductory brainwriting session and use the results later to build on further with other techniques.

Best Practice

If you were the facilitator, you'd brief ideation participants upfront on the problem statement, goals and important user insights from previous research and findings.

Encourage participants to jot down ideas on their idea cards for 3-5 minutes before passing on their ideas when you make the call.

Ideally, participants pass on idea cards 3-10 times depending on the problem statement and goals.

This all happens silently and without any interference or communication.

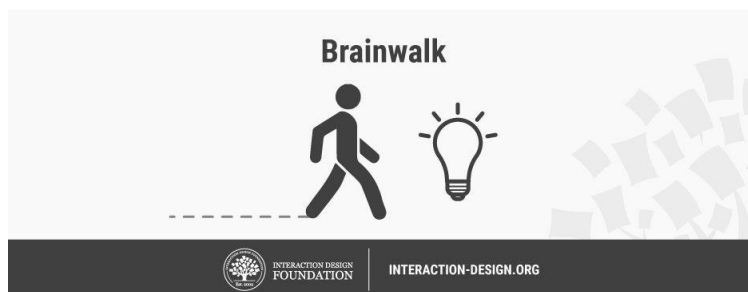
Encourage participants to push themselves for more ideas at least a couple of times, in the few minutes they have, in order to maximize the output and variation.

You should stoke the session with encouragement and provide questions or statements which push participants to think outside of their comfort zones.

The cycle can be repeated multiple times and can be applied to chunks of the problem being addressed, depending on the need.

After ending the cycle, each participant will briefly verbally present the thoughts on the idea card he/she ends up with by the end of the cycle to the rest of the team—in order to spark new streams of thinking or combinations of ideas. If you were the facilitator, you would often be taking notes on a white board.

When all team members have presented their idea cards, you can select the best ideas which you can continue to build and elaborate on in other ideation sessions. There are various methods you can use such as “Post-it Voting”, “Four Categories”, “Bingo Selection” and “Now Wow How Matrix”.

Brainwalking

Brainwalking is very similar to brainwriting with one small but highly impactful difference. Instead of passing idea cards or notes from one participant to another, participants have to get up from their seats and move to another spot around the brainstorming table or even to another table altogether. Bryan Mattimore came up with this technique; he's a specialist in the art of ideation and facilitation. Mattimore has many years and thousands of sessions of experience conducting ideation sessions. In his book Idea Stormers, he describes brainwalking as the: 'single best technique to use to begin an ideation session'.

Brainwalking gets people out of their seats, gets them moving, keeps energy levels up and mixes things up enough so that the group as a whole does not get stuck delving too long down the same one-way street, without a quick exit or U turn available. Brainwalking works best when it is easy to flip or to start from scratch.

Brainstorming is the most frequently practiced form of ideation. We recommend that you mix it with Brainwriting, Brainwalking and Braindumping and other ideation methods. Idea generation is an art form, which is dependent on appointing an experienced facilitator and having an experienced team

GAMESTORMING

Gamestorming is a set of ideation and problem-solving methods that are purposely gamified in order to dramatically increase levels of engagement, energy, and collaboration during group sessions. It involves some of the methods we've already mentioned, while adding gamification.

What is Gamification?

Gamification is a technique which designers use to insert gameplay elements in non-gaming settings, so they enhance user engagement with a product or service. By weaving suitably fun features such as leaderboards and badges into an existing system, designers tap users' intrinsic motivations so they enjoy using it more.

The Appeal of Gamification in User Experience (UX) Design

Gamification is a powerful tool to drive user engagement for several reasons. It's not about transforming user interfaces into games. Instead, you use it to inject fun elements into applications and systems that might otherwise lack immediacy or relevance for users

The Challenge for UX Designers

Gamification is notoriously difficult. You should strike a cautious balance between the “fun factor” and the tone of the subject matter. Moreover, you must tailor the gameplay and the rewards precisely to the users. Simultaneously, you must fulfill certain user needs if “players” are to use the system without forcing themselves to. Such needs include:

Autonomy – Users' actions must be voluntary; you shouldn't push them to adopt desired behaviors but instead insert subtle elements/prompts which they can find on their own and therefore feel in control of their experience.

Relatedness – Users need to feel that your brand cares about what matters to them. Customizing your design's content and tone to them are especially useful for fostering their loyalty.

Competence – Related to autonomy, this need is about keeping users comfortable as they discover your design by never feeling overwhelmed or confused. For example, as users typically don't like reading lots of text, you can use icons (e.g., a heart for “Save to Favorites”) or progressive disclosure.

Major Gamification Pitfalls:

Manipulation – Gamification is about motivating users by enabling them to have fun, not tricking them into doing things.

Building a Game – If you overdo the game features, you'll defeat the purpose of incentivizing users to complete real-world tasks.

Magic Paint – The system you're gamifying must be good per se. If it's subpar, gamification cannot make it a success.

Getting Gamification Right

To get gamification right, it's vital to know both the users and exactly how they should fulfill the goal or purpose of your gamified design, every step of the way. You also should understand that motivations will vary according to the task, objective and player. Likewise, the gamification mechanics must suit the users. So, when you choose to implement a leaderboard, points system, relationship-based approach, badges, etc., you should:

Enhance the experience from the users'/players' viewpoints by inserting the right gamification mechanics – First, do UX research to pinpoint who those “players” are and understand how they see their world. For instance, does their work environment suit a challenge-oriented experience with points, awards and competition? Or how about time constraints to complete the boring tasks and compulsory e-learning? Or might these seem adversarial and demotivate some users? Would a more discreet approach with elements of a journey of discovery and private personal achievement markers work best? From your research, you can create personas to help understand likely player types. From there, you can tailor a system that's right for all stakeholders involved.

Evaluate your design to monitor its effectiveness in bolstering user engagement – usability testing is invaluable here.

A FEW EXAMPLES OF GAMIFIED IDEATION SESSIONS INCLUDE:

Fishbowl: An ideation session in which participants sit in two circles, one smaller and one larger surrounding the smaller one. Participants in the inner circle discuss their ideas and brainstorm while participants in the outer circle listen, observe, and document the ideas and conversation points without saying anything. This forces some to listen and others to engage in brainstorming.

The Anti-Problem: The idea is based on flipping the problem. In the Anti-Problem is the opposite of the real problem that needs to be solved. In this session you seek to solve the anti-problem. This may provide inspiration that you could not have gotten access to by focusing purely on the real challenge, though it may generate ideas which are still related to the problem space. The ideas you generate can then be re-flipped to bring them back into the realm of the real problem.

Cover Story: This involves using a template that forces participants to create a cover story, including main image, headline, quotes, and sidebars with associated facts etc. It is a good method for vision generation sessions and helps create a cohesive picture of a broad subject area using the primary characteristics.

OBSERVATIONAL EMPATHY

What is empathy?

Empathy is the ability designers gain from research to understand users' problems, needs and desires fully so that they can design the best solutions for users. Designers strive for empathy by deeply probing users' worlds, to define their precise problems and then to ideate towards solutions that improve users' lives.

Empathy is the ability to understand and identify with another person's context, emotions, goals and motivations. In order to design great experiences, successful design firms actively search for empathic insights into their target group. In a design context, empathy serves a distinct purpose: to inspire design decisions in the early stages of the process. At IDEO, for example, the design team is so convinced about the positive effect it has on their projects that they actively advocate it to inspire other designers and innovators. Here, you'll learn how you can develop empathy for your target group.

Using empathy in the design process is on the one hand about collecting subjective information and on the other hand about objectively analysing it. The best way to collect the subjective information is to embed yourself in the context of your target group and gain personal insights into the experiences they have. There are three different approaches for you to use:

Looking at what people do

Asking people to participate

Trying things yourself

These should be used together to get empathic on an affective and a cognitive level.

Why is Empathy Important in a Design Process?

The aim of empathic design studies is not to seek solutions for recognized problems, but rather to look for design opportunities as well as develop a holistic understanding of the users. Design empathy is not only information and facts but also inspiration and food for ideas.

The Four Steps in Developing Empathy for your Target Group

As we mentioned before, there are three approaches to collecting the subjective information that you need so as to gain empathy for your target group. Each approach involves four general steps.

Discovery: enter the user's world and make contact with the user. This will help you get into the right mind-set to understand the user. Let's say you're designing a new workflow for employees working at a self-service food court to improve their efficiency. Maybe you have never exchanged more than a few words with the people behind the counters of a self-service food court. You don't know them. Walking around behind the scenes and getting a glimpse of the hours they put in and the limited space they have to move around in helps you get into the right mind-set. It triggers your designer's curiosity.

Immersion: wander around in the user's world to collect qualitative data. This helps you take the user's point of reference. When you start to collect data actively by participating as a member of the food court team,

talking to them during coffee breaks and taking pictures of things that stand out to you, you start to experience the context from your users' point of view.

Connection: resonate with the user, and recall your own experiences to connect and create meaning. This step may occur naturally while collecting the data. For example, when you find out how irritated the employees are by the lack of communication about changing menus and special offers, you might recall how it felt when you were in design school and teachers forgot to communicate clearly about changing mandatory literature for the next exam! Not having the right information to do your job properly may lead to a feeling of helplessness. You remember how it feels. You understand and identify with their context and feelings. You have empathic insights.

Detachment: step back into the role of designer, reflect and create ideas. While it may seem sufficient to get the empathic insights by following the previous steps, you need to look at your subjective data with a designer's mind so as to translate the empathic insights into ideas. A feeling of frustration about the lack of communication may seem solvable by actions directed at the team manager at first. Even so, after creating an overview of the insights and reflecting on it more objectively, you can use the informal communication that is already used between team members to create solutions that will give them a stronger feeling of control.



When you're designing a new workflow, for example for employees working at a self-service food court to improve their efficiency, you need to get empathic insights so as to understand who you're designing for. In the discovery step, you just wander around to trigger your curiosity. In the immersion step, you start to collect qualitative data in the user's context. In the connection step, you start to create meaning from the data you gather, by connecting them to your own experiences. You finally step back into the role of the designer to reflect and create ideas in the detachment step.

These steps reflect the balance you need between collecting subjective information on the one hand and objectively analysing it on the other. However, these steps are not terms that you are likely to use when you are trying to organise time and resources around the design research phase of a project. Rather than talking about immersion and detachment, you will be speaking in terms of collecting and analysing the qualitative data. Collecting information from your target group is the overlap between the immersion and connection steps. Analysis of the collected data is the detachment step.

The four steps in developing empathy for your target group are discovery, immersion, connection and detachment. In your day-to-day design process, you're more likely to refer to these steps as collecting and

analysing data.

How to Collect the Data You Need to Develop Empathy

Looking at what people really do in their current natural context or with prototypes you expose to them — This is a matter of observation of behaviours, interactions and products. Depending on the design assignment, you can plan to focus on certain aspects of the context. In any case, you will record your observations for further analysis and communication. You can do this by using notes, sketches of routes on maps, photographs, videos, etc. For example, mapping medical equipment and nurses on a hospital ward can lead to insights into how to improve the efficiency of workflows which your target audience will experience. And recording a video of walking a route through a government building at the eye-level height of a wheelchair user can lead to insights into how to improve the experience of accessibility.

Asking people to participate by recording their behaviour and context or by expressing their thoughts and feelings — You can do this without embedding yourself in the user's context by using a probes kit for context mapping (a collection of exercises designers give to a target group in a design project so as to obtain an understanding of their lives). However, embedding yourself in a context when asking people to participate may give deeper insights. For example, you might have seniors in a nursing home hand out cards with a pre-printed message which thanks the recipient, typically—in this case—a caregiver, an administration person or a member of the ancillary services (e.g., a janitor or cook). You could walk around with them when they give each one to a person they choose, thereby giving *you* enormous insights into the reasons behind their choices and the interactions these generate. Also here, you should use recording methods, such as filming or photographing, to preserve the data for later use.

Trying things yourself: to gain personal insights into the kinds of experiences others may have — This method is most time-consuming, but it may lead to different and more emotional insights. Resonating with the user on an emotional level may be easiest when you, for example, clean toilets in an office building for a few days and experience the mess people leave behind and the times people don't greet you like they do their colleagues. It is possibly the most difficult data to capture, but diary-style notes can be a good basis.

While all three approaches focus on gaining empathy, the first is more objective (more focused on cognitive aspects) and the last more subjective (more focused on affective aspects). Both are relevant in connecting with your users, so a mix of approaches will be most useful.

How to Analyse the Data Needed so as to Develop Empathy

The data you collect is very diverse, from personal impressions to pictures, quotes, maps and sketches. When analysing, your focus is on finding patterns or clusters that 'stand out' in some way as being an opportunity for design. For these patterns or clusters to emerge, you need to use your designer's intuition, but you can help it along with techniques borrowed from qualitative

research methods.

Best practice for an analysis process:

Gather all data into one room. Lay the materials out on tables and walls so you can see everything. Get two or three members of your design team together for the analysis.

Look at the data and start to see if some have similar themes. You can base these themes on underlying problems, recurrence of negative feelings, or anything else that sticks out and seems relevant to your design problem. Don't worry if you find it difficult at first. This step in the analysis process will get better with experience. Stick with it.

Label or cluster your data into categories, based on the themes you find. Use sticky notes and markers in different colours to indicate which pieces of data belong together.

Summarise the findings. At first, this could be an unorganised list of all conclusions. Then, you may add a hierarchy or (again) categories. Finally, you can summarise the findings in personas, requirements, mental models, scenarios, flowcharts or graphs. This will allow you to use the insights in any step of the remaining process and communicate them to stakeholders.

Empathy – The Glow You Put in Your Users' and Customers' Hearts

To understand your users/customers fully, you must see and feel their worlds from their perspective. And to access these core vantage points, you'll first need the right research methods. You want to gather reliable information from which you can distill your users' essences, as personas, to take forward in your design process. In user-centered design, user experience (UX) design and elsewhere, you need empathy. It even has a themed stage in the design thinking process: Empathize.

Your biggest challenge is to dig deep into your users'/customers' subconscious; they can't fully explain their precise needs. Designing for the human world is tricky, especially when users/customers access brands across many touchpoints and channels (e.g., online). In service design, ethnography is key to understanding their habits, motivations, pain points, values and whatever else influences what they think, feel, say and do on their user journeys. In ethnographic field studies, you observe what these users/customers do. Four methods are:

Shadowing – following users/customers around to get a day-in-the-life-of feel of what they experience.

Unstructured/Semi-structured Interviews – exploring hard-to-reach areas of their behavior in a naturalistic atmosphere, not systematically questioning them. This “hanging out” with them yields more honest, accurate insights. It's usually better to conduct semi-structured interviews,

strung loosely around an “areas-to-cover” framework in a discussion guide.

Diary Studies – letting users self-report. As with surveys, you rely on users to record things for you. Unlike surveys, diary studies help to capture “after-effects” over (typically) a 1-to-2-week period. Note: diary studies alone can’t reveal pain points effectively; they’re best combined with interviews.

Video Ethnography – video-recording enough material of participants in their environment as users/customers to gather insights about them.

How to Discover What Users Really Want
It’s best to remain informal and open-minded.

Here’s what to consider for an ethnographic study where you directly observe users interacting with a service (e.g., booking short-stay accommodation):

Introduction – Thank them and briefly explain your research’s purpose.

Context – Look around and note your users’ surroundings.

Note/observe/ask – Encourage them to continue their activities as though you weren’t there, letting you observe and ask as few questions as possible. When you do ask questions, ensure they’re open-ended and encourage more observations (e.g., “How?”).

Touchpoints & Channels – Pay attention to the touchpoints and service channels your users interact with (e.g., paying for room/property bookings by phone).

Tools – Note which tools these customers use throughout their journey.

Familiarity with Domain/Task – Note how comfortable they are with the various tools and tasks they use/perform.

Service Artifacts – Pay attention to the artifacts that are important throughout the service experience between the customers’ various touchpoints:

Physical items

Cognitive constructs (e.g., the customer’s changing understanding of the steps involved)

Social or emotional elements (e.g., hunting for a lockbox in an unfamiliar street)

Disconnects – Notice these, which happen anytime customers experience a problem with the service (e.g., they can't access the accommodation/property).

Needed ecosystem support – Watch for the points in the service where support from the backstage of the service is needed (e.g., the service-providing organization/agency must contact the landlord if the customer can't).

Wrap-up – Thank them at the end of the session and answer any relevant questions they have.

For Semi-structured interviews, order and ask your questions properly, stringing them loosely in a discussion guide featuring the following types of questions:

Introductory – e.g., “What was it like the last time you...?”

Follow-up – on what they've just said.

Probing – ask them to give an example/explain something.

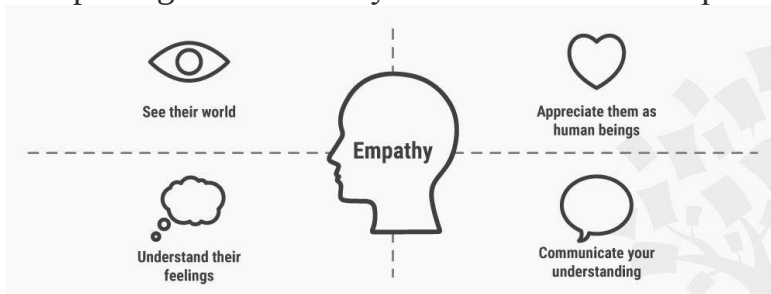
Specifying – if their descriptions are too general.

Direct – to introduce topics, etc.

Indirect – if you sense a direct question might lead the user, etc.

Structuring – to get back on-topic, etc.

Interpreting – to confirm you've understood the previous answer correctly.



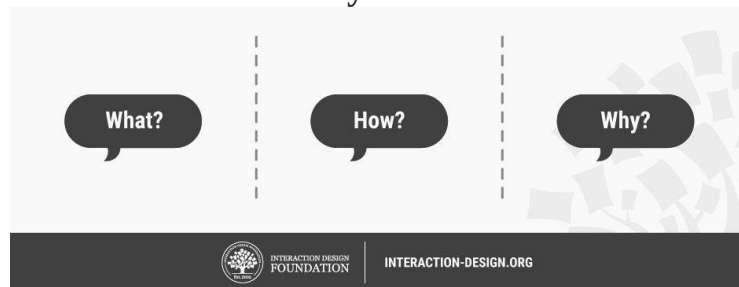
Developing Empathy towards People

The first stage (or mode) of the Design Thinking process involves developing a sense of empathy towards the people you are designing for, to gain insights into what they need, what they want, how they behave, feel, and think, and why they demonstrate such behaviors, feelings, and thoughts when interacting with products in a real-world setting.

To gain empathy towards people, we as design thinkers often observe them in their natural environment passively or engage with them in interviews. Also, as design thinkers, we should try to imagine ourselves in these users' environment, or stepping into their shoes as the saying goes, in order to gain a deeper understanding of their situations. In the following sections, we will outline some methods from d.school Bootcamp Bootleg that will allow you to gain empathy towards your users.

If we are to empathize with users, we should always try to adopt the mindset of a beginner. What this means is that, as designers (or design thinkers), we should always do our best to leave our own assumptions and experiences behind when making observations. Our life experiences create assumptions within us, which we use to explain and make sense of the world around us. However, this very process affects our ability to empathize in a real way with the people we observe. Since completely letting go of our assumptions is impossible (regardless of how much of a checkered reputation the word “assumption” has!), we should constantly and consciously remind ourselves to assume a beginner's mindset. It's helpful if you always remind yourself never to judge what you observe, but to question everything—even if you think you know the answer—and to really listen to what others are saying.

Ask What? How? Why?



UX designers' attitudes towards their work stem from natural curiosity, inquisitive behavior and constant critical appraisal of everything they encounter. Looking for the underlying factors and motives that drive users' behaviors and needs is what leads to successful design.

By asking the three questions — What? How? Why? — we can move from concrete observations that are free from assumptions to more abstract motivations driving the actions we have observed. During our observations, for instance, we might find separately recording the “Whats”, “Hows” and “Whys” of a person's single observation helpful.

In “What”, we record the details (not assumptions) of what has happened. In “How”, we analyze how the person is doing what he/she is doing (is he/she exerting a lot of effort? Is that individual smiling or frowning?). Finally, in “Why”, we make educated guesses regarding the person's motivations and emotions. These motivations we can then test with users.

Photo and Video User-based Studies

Photographing or recording target users, like other empathizing methods, can help you uncover needs that people have which they may or may not be aware of. It can help guide your innovation efforts, identify the right end users to design for, and discover emotions that guide behaviors.

Personal Photo and Video Journals

In this method, you hand over the camera to your users and give them instructions, namely to take pictures of or video-record their activities during a specified period. The advantage is that you don't interfere or disturb the users with your personal presence, even though they will adapt and change their normal behavior slightly as they know that you'll watch the video or see the photo journal later. In a similar way to using personas, by engaging real people, as designers we gain invaluable personal experiences and stories that keep the human aspect of design firmly in mind throughout the whole process.

Interviews



Interviews are an important part of the UX designer's skillset for empathizing with users. However, an interview will yield only minimal results if you are not prepared to conduct it with genuine empathy.

One-on-one interviews can be a productive way to connect with real people and gain insights. Talking directly to the people you're designing for may be the best way to understand needs, hopes, desires and goals. The benefits are similar to video- and camera-based studies, but interviews are generally structured, and interviewers will typically have a set of questions they wish to ask their interviewees. Interviews, therefore, offer the personal intimacy and directness of other observation methods, while allowing the design team to target specific areas of information to direct the Design Thinking process.

Engaging with Extreme Users



Extreme users are few in number, but it doesn't mean you should disregard them and aim just for the main bulk of users instead. In fact, they can provide excellent insights that other users may simply be unprepared to disclose.

By focusing on the extremes, you will find that the problems, needs and methods of solving problems become magnified. First, you must identify the extremes of your potential user base; then, you should engage with this group to establish their feelings, thoughts and behaviors, and then look at the needs you might find in all users. Consider what makes a user extreme and you'll tend to notice it's the circumstances involved. A basic example is a grocery store shopping cart and a shopper with five very young children in tow – there are two fold-down seats in the cart, but the other kids (who are also too young to walk) must go somewhere. Our shopper is, therefore, an extreme user of the shopping cart design.

On the one hand, if you can manage to please an extreme user, you should certainly be able to keep your main body of users happy. On the other hand, it is important to note that the purpose of engaging with extreme users is not to develop solutions for those users, but to sieve out problems that mainstream users might have trouble voicing; however, in many cases, the needs of extreme users tend to overlap with the needs of the majority of the population. So, while you may not be able to keep everyone happy at all times with your design, you can certainly improve the chances that it will not frustrate users.

Analogous Empathy

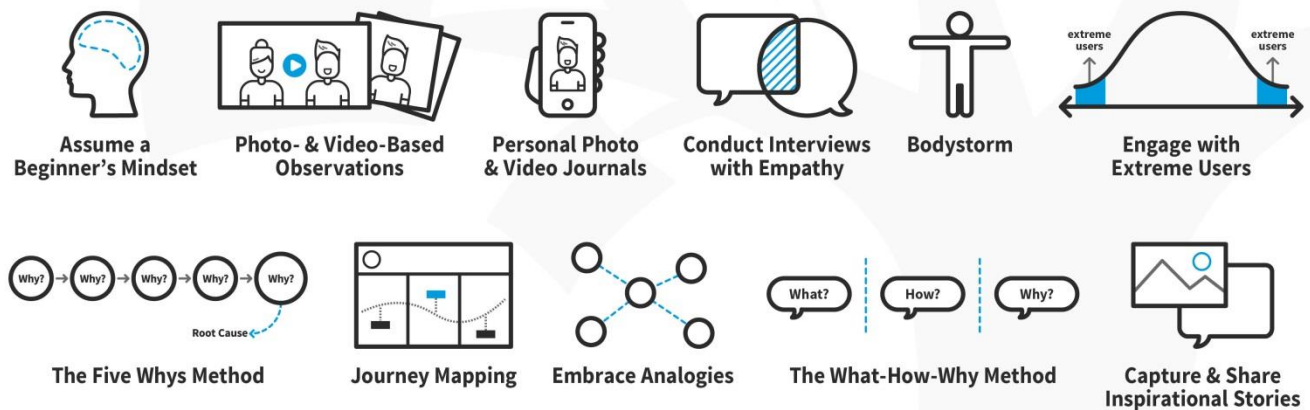
Using analogies can help the design team to develop new insights. By comparing one domain with another, we as designers can conjure different solutions that would not necessarily come to mind when working within the constraints of one discipline. For example, the highly stressful and time-sensitive procedure of operating on a patient in a hospital emergency room might be analogous to the process of refuelling and replacing the tires of a race car in a pit stop. Some of the methods you might use in analogous empathy include comparing your problem and another in a different field, creating an 'inspiration board' with notes and pictures, and focusing on similar aspects between multiple areas.

Sharing Inspiring Stories



In the words of the great author Terry Pratchett, “People think that stories are shaped by people. In fact, it's the other way around.” We might paraphrase slightly here, as it's true that products are shaped by the stories that people tell about them.

Empathize Methods



Interaction Design Foundation
interaction-design.org

What's good to note is this empathic phase of design thinking is named differently depending on which version of the methodology you follow. Different schools and companies which use design thinking have called empathic research "the Empathize stage" (as we do), "the understand phase," "the hear phase," and simply "looking," as well as a number of other terms. Regardless of which phrase you're familiar with, the core is essentially the same — empathy is deeply human-centric and is essential at the start of any design process.

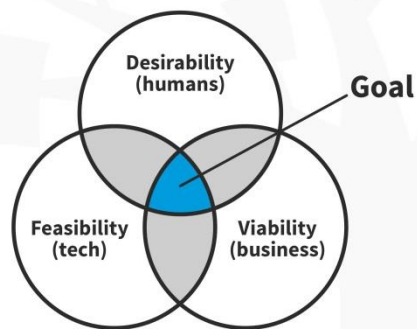
Empathy is Crucial to Business Success

Empathy can also be deemed an essential component of business solutions when you look at things from the perspective of profit. You may create solutions which completely miss the mark if you develop solutions in isolation — you need to gain essential insights about your users if you want to remain relevant in the market.

Many leaders within the fields of innovation, learning and entrepreneurship have pointed to three key parameters which define a successful product or service: desirability, feasibility and viability.

It's not enough that the technology exists (i.e., feasibility is present) and that profits or business benefits may be derived (i.e., it is viable). Users need to feel a sense of desirability towards a

solution. We can only design a desirable product or service when people's needs, experiences, wants and preferences are properly understood.



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One clear example of this is the iPod. Many MP3 players came and went throughout the late 1990s, and didn't create much of an impact. Then, along came the iPod in 2001. It not only provided a technological solution, but also delivered a completely desirable and viable experience. This meant Apple took the lead in the market and continued to hold it for many years to come, generating huge profits along the way.

Empathy Helps You Read Between the Lines

Empathy is also the only way to thoroughly understand what people mean, rather than just absorb what they say. You'll have noticed by now people do not necessarily always cover the details when they share stories and other information.